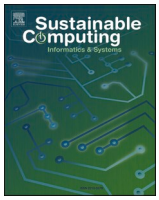
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Komodo Mlipir Algorithm-based optimal route determination mechanism for improving Quality of Service in Vehicular ad hoc network

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ABSTRACT

In Vehicular ad hoc network (VANETs), Quality of Service (QoS)- aware protocols helps in handling the necessitated demand of delay sensitive applications for facilitating intelligent transportation. The fundamental challenge of VANETs lies in the process of establishing vehicle to infrastructure and vehicle-to-vehicle communication that are prone to link failure. Bio-inspired algorithms are identified to provide reliable solutions for securing the links of VANETs. In this paper, Komodo Mlipir Algorithm-based Optimal Route Determination Mechanism (KMAORDM) is proposed for achieving best optimal route that guarantees enhanced QoS. This QoS aware routing protocol utilizes the exceptional potentiality of Komodo Mlipir Algorithm attributed in terms of exploration and exploitation for the objective of evaluating and memorizing the impactful factors that helps in exchanging the necessitated messages to its neighbours. It adopts Mlipir as a best mode of modelling the behaviours of vehicular nodes during the process of data routing in VANETs. It uses a cache strategy which discovered reliable traversed routing paths such that the pre-cached route is used for data transmission. This cache strategy paves the way for preventing the route identification and maintenance process with minimized routing overhead. It also identifies the fittest vehicular node from its one hop distance as the successive forwarder in the absence of pre-cached route for addressing link failure during reliable packet transmission. The simulation results of this proposed KMAORDM approach with different vehicular nodes confirmed minimized communication overhead of 19.32 %, reduced network latency of 18.64 %, maximized throughput of 18.54 % better than benchmarked approaches.

1. Introduction

Vehicular ad hoc networks (VANETs) represent the advanced form of mobile ad hoc networks (MANETs) which is completely dedicated for attaining wireless connectivity between vehicles. In intelligent transportation system, the vehicle communicates either by using single-hop or multi-hop communication for the objective transmitting information from the source to the destination vehicular nodes [1]. The application of VANETs mainly target on the objective of enhancing efficient path planning, emergency warning, congestion control, commuter safety with reduced time of travel. In specific, control packets are used by the vehicular nodes for exchanging the status of traffic such that the information related to congestion and neighbourhood accidents are made aware. The environment of VANET need to identify false information generated by attackers for the purpose of achieving drivers' safety [2]. Moreover, the vehicular network with maximized security increases the possibility of guaranteeing reduced consumption of fuel and human causality which in turn helps in managing significant improvement in the system of transportation. In this context, roadside units (RSUs) represent the most potential infrastructural backbone of ITS for establishing the above-mentioned objectives. These RSUs during data transmission facilitate the process of information exchange between the vehicles that mounted application unit and On-Board Unit (OBUs) for provisioning safety applications. The application units may communicate with the OBUs in a wireless or wired manner, such that the communication of vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V) is achieved [3]. However, the transmission of data can happen through the process of broadcasting, unicasting and multicasting depending on the application scenarios. VANETs generally generates and exchanges control packets and data packets depending on the process of broadcasting, but they face the challenge of broadcast storemen which inevitably incurred unnecessary bandwidth. The successful control packets transmission in VANETs is generally achieved using route discovery process. Moreover, small sized control packets cannot

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guarantee successful data packets delivery [4]. On the other hand, the routing protocol possesses the potential of handling the hybrid topologies since the communication environment is heterogeneous in nature [5]. When the route determined during data exchanges are not globally optimal, then it is not efficient for the routing protocols to relay on the determination of the local best route for the transmission of data. Further, the path of the vehicular nodes can get overlapped when they do not possess the knowledge related to the neighbour nodes' path. In addition, the implementation of an efficient routing protocol is more crucial since exchange of information with reliability is more important for achieving successful ITS. In precise, increase packet delivery ratio and minimized network latency guarantee Quality of Service (QoS) since road accidents can be controlled by warning about the road conditions in the vehicular network [6]. The probability of link failure is very high since the highly mobile vehicular network makes the routing in VANET more difficult. At this juncture, cache-supported RSUs and OBUs indicates the effective way for achieve potential data dissemination in a more precise and safe manner. The cache memory of RSUs collects the information generated by OBUs for transmitting them to other OBUs and RSUs using the benefits of back-haul communication [7]. But storing route information in RSUs is identified to be more effective than OBUs since the RSUs caching capability is comparatively greater than the caching capability of OBUs. RSUs are used during V2I communication for facilitating the vehicular nodes to retrieve information which are required during message exchange. This methodology of utilizing cached route information from RSUs speeds up the routing and incurs minimized network overhead in V2V communication [8].

In specific, vehicular clustering is identified to address the problems related to high mobility of vehicles that crumbles QoS, hidden terminal problems, broadcast storms and scalability in VANETs. The vehicular nodes organize themselves for the objective of constructing clusters depending on the factors of position, number of vehicles with the range of transmission, direction, speed and vehicular speed into account. Each of the constructed cluster possesses a cluster head (CH) which plays an indispensable role in controlling the cluster and cluster members (CMs). The impact introduced by most of the clustering algorithms gets declined with increase in the number of vehicles in the urban environment. Cluster stability is determined to be minimized due to the high mobility rate of vehicular nodes. Hence a specialized clustering scheme that considers the mobility-related factors for preventing cluster partitioning frequently is essential. The adaptable properties offered by the bio-inspired algorithms is confirmed to be potential in handling the major issues that are concerned with weakness of the communication link in the urban environment.

In general, routing protocols are classified into geographic and topology-based proposals, in which the earlier one adopted location-based services for achieving data packets routing in the network while the latter utilized the information related to network topology for connecting vehicular odes with maximized reliability. The classical topological protocols implemented for MANETs were included into VANET with adaptability due to their self-organizing and distributed characteristics [9]. However, these classical topology-based protocols are not ideal when the environment of VANET need to handle the issue of rapid node mobility. The primary step of topology-based protocols mainly targets on discovering the topological route from the source to destination vehicles before data transmission. These protocols can be either proactive or reactive depending on the mechanism adopted for route construction. The proactive routing protocols are complete table-based, and the vehicular nodes need to transmit the help packets in regular intervals for discovering the neighboring nodes [10]. On the other hand, reacting routing protocols help in detecting routes on demand i.e., whenever a packet from the source to the destination vehicular nodes need to be forwarded or sent. Some of the instances of proactive and reactive protocols are Dynamic Source Routing (DSR), Ad-Hoc On-Demand Routing (AODV), Destination-Sequenced Distance Vector (DSDV), Optimized Link State Routing (OLSR). Among the proactive and reactive protocols used in VANETs scenario, researchers mainly used AODV protocol by enhancing its application [11]. Some of modifications done over AODV protocol for making it highly suitable to application scenarios are ad-hoc on-demand multipath distance vector (AOMDV), priority ad-hoc on-demand multipath distance vector (P-AOMDV), Greedy perimeter stateless routing (GPSR), road aware geographical routing protocol (RAGR), etc. In specific, AODMV and P-AODMV concentrated on the process of determining multiple routes during route discovery at every time, and provision clear vision of the network. Then GPSR and RAGR protocols are location-based and, are most preferable since they utilized the vehicles' geographical position into account during data transmission [12]. Most of the location-based routing protocols completely depends on RSUs installed at each road intersection, and this process of configuration makes the task more expensive. The topology-based routing protocols determines the path of routing depending on the topological structure underlying in the network. These topology-based routing protocols concentrates on attaining accurate path to the destination using shortest paths with minimized rate of packet dropping [13]. However, unreliable routes discovery, communication overhead and increased delay increases the probability of network performance degradation during the utilization of topology-based routing protocols. But the protocols with geographical routing phenomenon include location-based services for identifying the potential path between the source and destination nodes during the data transmission process [43]. Hence geographical routing protocols improves the probability of decreasing packet dropping with enhanced route reliability. But these geographical routing protocols still has the routing concern with respect to end-to-end delay [44]. Moreover, the above-mentioned routing protocols are associated with the classical methods of computation, and hence confirmed to be complex, less practical and not suitable for extensive vehicular networks.

From the recent past, bio-inspired routing protocols has attracted more attention of research in vehicular networks due to their potential properties of robustness, adaptability, ad-hoc nature, and scalability. Bio-inspired routing protocols are highly preferable for large vehicular network because they mimic the foraging, hunting or mating behavior of animals or swarms for performing local and global search such that it aids in determining the optimal route between the source and destination vehicular nodes. These bio-inspired routing protocols are the ideal selection as they are capable in handling the dynamic changes in node behaviors and topology in the case of vehicular network [14]. The problem of routing the packets from the source to the destination nodes in VANETs represents a NP-Hard problem. Some of predominant bio-inspired routing protocols used for efficient data routing in vehicular network includes ant colony optimization (ACO), Particle Swarm optimization (PSO), Jaya algorithm, Grey Wolf Optimization Algorithm, canine olfactory algorithm, etc. Especially in an urban VANET environment, the successful delivery of packets is always established between the source and destination vehicles only when packet forwarded is present. This requirement of urban VANETs introduces the network overhead when the control packets are broadcasted by all the vehicles in the network. This issue of urban VANETs is also addressed in the proposed routing protocol [15].

The importance of Bio-Inspired Algorithms for VANETs is adopted for enhancing the frameworks of Intelligent Transport Systems (ITS) is mainly due to the following reasons that includes, i) it is capable of effectively addressing the topological structures variations visualized in VANETs network dur to its potentially in adapting its factors with respect to different scenarios considered for implementation and self-organization significance, ii) they are more accurate in the process of identifying the damaged nodes of the network since they help in incorporating maximized level of trade-off between the phases of exploration and exploitation, iii) its potentiality in security improvement which aided in facilitating practical approaches that attributed towards the minimization of security attacks in VANETs, iv) It introduced low complexity in the process of addressing the computational

issues included in VANETs implementation, and v) It helped in reducing the reducing delay and network overhead with improved factor of convergence and packet delivery ratio.

In this paper, Komodo Mlipir-Based Route-Finding Algorithm (KMRFA) is proposed for managing the objective of determining the route between the source and destination vehicles in VANETs which is an integral part of ITS. This KMRFA algorithm is a bio or nature-inspired approach which mimics the exceptional capability of Komodo Mlipir in terms of foraging and reproduction. This KMRFA algorithm is developed based on the inspiration of the populations' self-adaptation which helped in better control of balance between the degree of exploitation and exploration. This algorithm comprises of three main phases in which the first phase, better solution is determined through low exploration and high exploitation. In the second phase, the algorithm determines better solution either by using exploration or exploitation. This algorithm in the third phase diversifies the candidate solutions through the Mlipir movement for attaining maximized global solutions in the search space. This KMRFA algorithm determines the optimal route and the high fitness member used for packet forwarding through the inclusion of Mlipir coefficient. The performance of this KMRFA algorithm is evaluated using packet delivery rate, network latency, throughput and communication overhead with different vehicular density and speed of vehicles.

Major Contributions of the proposed work.

The major contribution of the proposed work used for establishing reliable routing is listed as follows.

i) Exploration and exploitation capability of Komodo Mlipir-Based Route-Finding Algorithm (KMRFA) is used for determining the optimal route between the source and destination vehicle such that maximized packet delivery with minimized end-to-end delay is achieved.
ii) It is proposed with the potentiality of reducing the communication overhead in urban VANETs through the incorporation of cache memory-assisted RSUs such it can establish reliable communication with the other neighbourhood RSUs through the backhaul link.
iii) It is proposed with the well-balanced rate of exploration and exploitation of KMRFA algorithm such that neighbourhood RSUs can facilitate the previously established route information stored in them when the vehicular clusters; movement occurs in a prediction and uniform fashion.
iv) It facilitates limited control packet broadcasting when the previous route is not existing the cache memory of RSUs such that expenditure in route maintenance, delay in route discovery and link failure can be significantly reduced.
v) It is experimental validated and compared with the other bio-inspired, topological, and geographical routing protocols in terms of packet delivery ratio, throughput, routing overhead and network latency with different speed of vehicles, data packet size, distance between the source and destination, vehicular traffic density.

The remaining section of the paper is structured as follows. Section 2 presents the comprehensive review of the existing bio-inspired routing protocols contributed to the literature over the recent years with the pros and cons. Section 3 presents the detailed view of the proposed KMRFA algorithm with the system model, system of operation and mathematical adopted for determining the optimal route between the source and destination vehicle. Section 4 demonstrates the performance evaluation of the KMRFA conducted using packet delivery rate, network latency, throughput and communication overhead with different vehicular density and speed of vehicles. Section 5 finally draws the conclusion and future scope of enhancement related to the proposed research.

2. Related work

In this section, the comprehensive review of the existing bio-inspired routing protocols contributed to the literature over the recent years with the pros and cons.

Initially, Rajput et al. [16] developed and implemented a hybrid swarm intelligent routing algorithm that concentrated on provisioning QoS and determining the path from the source to the destination vehicular nodes. It was proposed as an optimization problem of routing which determined optimal solution with the modification of ACO, PSO, Cuckoo Search and Firefly algorithms. It was proposed as a reliable attempt to solve multi-constraint QoS optimal path problem with a potentially developed cost function. It is the first approach which explored the experimental and theoretical significance of the adopted optimization algorithm towards better QoS achievement. It was proposed with the well-balanced capability of exploration and exploitation which helped in determined feasible solutions from the search space, such that optimal route can be established with better optimality. The results of this HSIRA also confirmed its predominance in improved throughput, packet forwarding potential, cluster construction with reduced communication overhead and end-to-end delay. Ramasamy et al. [17] proposed a swam intelligent routing algorithm using Fuzzy-based PSO (FPSOA) for preventing collision and excessive overhead during the process of data delivery in VANETs. This FPSOA algorithm provisioned a high-quality route for communication between the vehicular nodes through the inclusion of two steps such as selection for candidate nodes and construction of optimized routing model. It adopted the objective function that targeted on end-to-end minimization and packet delivery rate improvement in the first step using the merits of fuzzy logic. It also adopted PSO in the second step for establishing the optimal route through a position-based approach. The results achieved for FPSOA algorithm confirmed better packet delivery rate with minimized execution time and end-to-end delay under different vehicular densities and speed.

Ramamoorthy and Thangavelu [18] proposed an Enhanced Hybrid Ant Colony Optimization Routing Protocol (EHACORP) for preventing low throughout, packet delivery rate and minimized packet loss rate, frequent disconnections, and high communication overhead during the determination of lengthy route established between the source and destination nodes in VANETs. This EHACORP protocol used the shortest path for improving the efficiency of the routing mechanism. This protocol determined the shortest path depending on the factors that included minimized number of hops between source and destination nodes and reduced communication costs. It incorporated two main phases, in which the first phase is responsible for computing the distance between vehicles and the second phase focuses on the adoption of source-based ACO which helped in guiding the search agents to construct shortest path using the distance factor. The simulation results of EHACORP protocol confirmed better throughput with minimized packet loss rate and routing overhead on par with the compared AODV and F-ANT algorithms evaluated under different vehicular speed and vehicular density. Attia et al. [19] contributed an Advanced Greedy Hybrid Bio-Inspired (AGHBI) routing protocol for facilitating a smart and effective smart future transportation system. This AGHBL routing mechanism was developed a greedy approach which concentrated towards the improvement of network performance, It specifically utilized the benefits of bee colony optimization for maintaining the path with reduced workflow and selecting the optimized route with maximized QoS. The simulation results of AGHBL routing mechanism confirmed acceptable overhead and minimized hop count between all the vehicles with improved packet delivery rate under the impact of V2I and V2V communication.

Further, Kumbhar et al. [20] proposed a routing protocol named UniFaVa which is a unified framework useful for facilitating routing decision that confirmed maximized rate of data transmission. This UniFaVa used vehicular nodes with sufficient residual energy as the

relay. It analytically predicted the time of connectivity for ensuring reliable data transmission during V2V mode of communication. It specifically adopted the hypervolume-based multi-objective optimal routing method for achieving analytical exploration of the data packets exchanged in the network. The results of UniFaVa determined using the OpenStreetMap related to the real traffic traces of New York City and Seoul with SUMO confirmed minimized connectivity time of approximately 5 s for establishing multi-hop reliable routes. The results also confirmed 90 % of packet delivery rate with one extra hop as the short hop distanced used for achieving reliable data propagation. Ramamoorthy and Thangavelu [21] proposed another energy-efficient shortest path without congestion (EESPWC) algorithm for maximizing throughput during data dissemination in VANETs. This EESPWC algorithm was proposed as a reliable routing algorithm that targeted on the satisfaction of the parameters that included reduced energy consumption, low packet loss ratio, low end-to-end delay, low routing overhead and increase throughput. It specially used ACO with the fitness parameters of residual energy and improved distance for determining congestion in the network. It included the merits of roulette wheel for selecting the best optimal path from the source to the destination using the fitness values associated with the link established between vehicular nodes. This improved ACO model was developed by relying on the parameters of congestion levels, distance, and residual energy for computing the degree of pheromone ranges. It identified the energy-efficient shortest route without congestion through the integration of links that possesses high fitness values. The simulation results of EESPWC algorithm confirmed minimized packet loss rate, end to end delay and routing overhead with maximized packet delivery rate and throughput compared to the ACO variants-based routing algorithms.

Ramamoorthy and Thangavelu [22] again proposed another enhanced ACO (EACO) algorithm that determines the optimal routes with maximized network lifetime for achieve reliable communication. This EACO algorithm identifies the optimal routes with minimal hops depending on the factors of evaporation rate, hop count, received signal strength, and distance. It mainly concentrated on the links that possesses minimum hops through the identification of high level of connectivity and selecting path with shortest distance established between the vehicular nodes. It was proposed with the capability of selecting shortest paths using minimum hops such that the option of improving route lifetime through high connectivity and reducing the possibility of frequent link breaks between the vehicular nodes is achieved. The results of EACO confirmed better success rate during route discovery process under the evaluation with different vehicular density and speed. Its performance was also identified to be significant in improving throughput, packet forwarding rate and packet latency independent of the changes in received signal strength evaluated using speed and density of vehicles. Then Seth et al. [23] implemented a lane-based advanced forwarding protocol (LBAFP) for addressing the factor of nodes mobility which introduces routing challenges and inconsistent network coverage in vehicular networks. This LBAFP protocol uses the vehicular nodes' position coordinates for identifying the destination zone for facilitating packets' broadcasting towards the destination direction. It used the broadcasting method of suppression for minimizing the network routing overhead such that interferences that arises with respect to various road segments are completely minimized. This protocol reduced the overhead through the generation of classical beacon forwarding mechanism that employed the benefits of suppression, broadcasting, and greedy forwarding strategy into account. It adopted the greedy method for recovering from the stage of local maximum through a method of carry-and-forward. The simulation results of LBAFP protocol helped in realizing its potentiality in minimizing latency with superior packet delivery rate in congested networks.

In addition, Nahar and Das [24] proposed a Meta-Heuristic GWO--based Reinforcement Learning method (GWO-MHRLM) for determining the optimal routes that maximizes network throughput in VANETs. This GWO-MHRLM possessed the merits of global search which in turn inherited parameterized strategy which helped in eliminating uncertainty in rewards and exploration of vehicular nodes' state that helps in optimizing the structure of multilevel network. It incorporated the balanced rate of exploration and exploitation using GWO which aided in determining and achieving optimal solution through Temporal Difference Learning. This adoption of Temporal Difference Learning enforces the cluster heads to acquire knowledge about how route request is generated and exchanged using the parameters of QoS. It also learned and flexible adapted the actions of successive moves of routing depending on the input received from each vehicle during the previous computation or assessment phases. Finally, Ghosh et al. [25] contributed a Canine Olfactory Route-Finding Algorithm (CORFA) with improved QoS for achieving the best feasible route with reduced packets transmission in VANETs. This CORFA algorithm adopted the canine's exceptional potential for acquiring knowledge about the environment such that message can be effectively exchanged between the neighbouring nodes. It utilized the canine's behavior as the better selection for modelling their behavior in the process of VANET routing. It used cache content of RSUs for exchanging messages about the previous stage during the process of routing between vehicular nodes. It reduced routing overhead by preventing the procedure of route discovery and maintenance through the incorporated of cache-supported RSUs. It considered fittest vehicles with each hop as the successive forwarder of data transmission under the absence of pre-cached route. It also addressed the scenario of link failure by forming multiple routes during route discovery. The results of CORFA confirmed minimized overhead and network latency with improved throughput and packet forwarding proportion independent to the speed and density of vehicles.

The limitations identified from the literature review [31–42] motivated the option of formulating and implementing a bio-inspired optimized route determination algorithm which could help in better stable network with energy efficiency.

3. Komodo Mlipir-Based Route-Finding Algorithm (KMRFA)

In this section, the system model, principle of operation with respect to RSU caching and Fittest Neighbour Selection, mathematical modelling of KMRFA and Background of Komodo Mlipir Algorithm (KMA) is presented in detail.

This proposed KMAORDM approach is developed as a route-determining algorithm which helps in managing the route-finding process in VANETs which is considered as the ITS integral part that is also use for transmitting the data packets. This proposed KMAORDM approach is implemented as a metaheuristic and nature-inspired approach that included the exception ability of movement exhibited by small male Komodo dragons for protecting themselves from large male Komodo dragons, and hunting or foraging the leftovers of the large male population. This optimization phenomenon is adopted for routing process improvement which comprises of exploration phase and exploitation phases. The phases of exploration and exploitation is responsible for determining the globally searching the leftovers that remains after the prey hunting achieved by large male population of Komodo dragon, and the behavior of Mlipir they exhibit when they prevent themselves from the large population of Komodo dragons. In this approach, the fittest member either the large male or small weaker male population determined depending on the fitness value is considered for facilitating the process of data routing in vehicular nodes. In this case, the vehicular nodes (wither large males or small weak males) with maximized fitness value is used or achieving the functionality of packet forwarding.

In this proposed KMRFA algorithm, the cache memory of RSUs is used for collecting the information generated by OBUs for transmitting them to other OBUs, such that they fully utilized the merits of back-haul communication. This storage of route information in RSUs is identified to be more effective than OBUs since the RSUs caching capability is comparatively greater than the caching capability of OBUs. This caching

strategy helped in discovering reliable paths of routing which prevents routing overhead rather than increasing the possibility of introducing a greater number of packets retransmission. In specific, RSUs are used during V2I communication for facilitating the vehicular nodes to retrieve information which are required during message exchange. This methodology of utilizing cached route information from RSUs speeds up the routing and incurs minimized network overhead in V2V communication.

3.1. System model

Dissemination of data plays a significant role in dealing with dynamic network topology. Road Side Units (RSUs) deployed in a particular area are efficient in managing data [26]. Vehicles include On-Board Units (OBUs) which are used for communicating with RSUs. RSUs support caching, and hence involve less routing overhead and offer improved performance. Every RSU includes a cache with capacity 'C' covering an area of size 'A'. It stores the routing information for a particular period. Newly arriving vehicles demand for routing information as they enter a particular region. There are chances for vehicles travelling along the same roadways in same direction to demand for the same contents offered by RSUs. In case the RSU has the essential routing table, there is no need for route discovery. Packets can be forwarded without routing overhead, thus reducing the involved delay. The source which desires to send packet to the destination, broadcasts sniffing packet with routing demands (small control packet). The corresponding RSU forwards cached content which recalls successful transmission paths. The source ensures path availability and forwards data through that path. When route is unavailable, route discovery becomes essential. The source broadcasts barking packet (control packet) along with Time and Link Attribute (TLA). The vehicles positioned in the next hop receive and save it with corresponding timestamp. Whenever nodes receive the broadcast packets, they are not broadcast further. When a node receives a packet which is available in TLA, it forwards it to the corresponding subsequent neighbour. Once the packet reaches the destination with complete TLA information, it chooses the best path for transmitting data and forwards to source. As routing information includes a timestamp of every node, the best node in every slot is chosen. Several paths are identified, thus facilitating the selection of next better path when link or better path is unavailable. During data transmission, the chosen best route may be absent, and hence the next best path is chosen.

Assumptions.
Following are the assumptions made.

- Every RSU has similar caching as well as coverage ability.
- RSUs are evenly positioned in the region considered for investigation.
- Every vehicle has an OBU and Global Positioning System (GPS)
- Vehicles are connected using Dedicated Short-Range Communication (DSRC) protocol in communication range of 150 m.
- Geographic position, direction along with velocity are accessible.
- Beacon packets are employed for transmitting information about network and nodes obtain packets from source at varying times.

3.2. Principle of operation

Vehicle locations are got using location-based services. When a source desires to send packet to destination, it checks whether the destination lies within coverage. In case it is found to be within coverage, it forwards the data to the target directly. If not found, it checks for route availability. To establish paths immediately, it sends query to neighbouring RSU The caching method used for receiving routing table is detailed in the ensuing algorithm. It uses binary variables to ensure availability of connection and routing content. RSU forwards routing information for data transmission to take place. If path is unavailable, process of finding fittest neighbour commences. Route discovery is essential for efficiently routing data. Packets from vehicles within the range are only considered. On receiving a barking packet from previous hop, nodes broadcast them to the ensuing hop. Next hop nodes are chosen based on link quality and timestamp. When a barking packet is received from same hop count, broadcasting is stopped. This continues until the destination is reached.

Algorithm. **RSU caching**.

Input: Vehicle within communication range at 't', Quantity of RSUs
Output: Routing table that is pre-cached
Get request for connection from source.
Check '$\varphi_{V,R}$', '$\delta_{V,R}$'
while (1)
    Get routing tables to form paths for data dissemination.
    if (route information is completely obtained)
        RSU obtains Acknowledgement.
    end
    if (Cache of RSU is completely filled)
        Reject the list in the cache.
    end
end

After receiving the complete routing information within permitted delay, the destination chooses the fittest member for every hop (refer algorithm). Barking co-efficient is built by considering quality of link along with probability of successful packet reception. It verifies every vehicle from hop count. Nevertheless, as the topology is dynamic, paths are not available while transmitting packets. As information about paths are available at destination, alternative best paths can be chosen in case of failure of links.

Algorithm. Fittest Neighbour Selection.

Input: Iterations $[H_c, H_{Dest}]$, Set of vehicles $[v_1, v_2, \ldots, v_N]$
Output: $V_{High}(B_c^v)$
Set solution space $(Vehicle_{pos,vel})$
while $(H_c < H_{Dest})$
    for (i = 1 to n) do
        Assess 'B_c' for current hop.
        Order vehicles and choose best solution for sending packets.
    end
end

Mathematical Modelling.

To efficiently design a routing algorithm, searching technique for determining fittest member from the search space is to be modelled mathematically. The algorithm functions in 2 stages.

- ✓ **Phase 1:** Check whether search space is visited already or not.
- ✓ **Phase 2:** If not visited, explore search space to find the fittest vehicle for forwarding messages

Phase 1

1. RSUs ($R = R_1, R_2 \ldots R_q$) are arranged by placing them at intersections. They are linked using high-speed LAN. They have computational as well as caching capabilities such that vehicles can reclaim essential details from respective RSU. RSUs with same range of transmission as well as storage capacity are placed uniformly. The overlapping area of 2 successive RSUs is less. Vehicles establish communication with an RSU at a time. Timeframes are separated in distinct time slots ($t = 0, 1, \ldots t-1$). Each time slot involves only one variable. As requests are sent using control packets, it sends cached content using Vehicle-to-Infrastructure (V2I) links [27].

$$CR_{V,R}^t = BW_{V,R}^t \times \log_2\left(1 + \frac{P \times G_{V,R}}{N_0 \times BW_{V,R}}\right)^t \quad (1)$$

Where,
$CR_{V,R}^t$ - Rate of communication between 'V' and RSU 'R_i'
P - Power of transmission
$G_{V,R}$ - Channel gain
$BW_{V,R}$ - Channel bandwidth between 'V' and 'R_i'
N_0 - Noise power density

2. In Eq. (2), binary variable $(\varphi_{V,R})$ is employed for determining status of connection between 'V' and 'R_i'. In Eq. (3), '$\delta_{V,R}$' represents the availability of routing content. At time '$t \in T$',

$$\varphi_{V,R} = \begin{cases} 1, \text{V' and 'R}_i\text{' are linked} \\ 0, \text{otherwise} \end{cases} \quad (2)$$

$$\delta_{V,R} = \begin{cases} 1, \text{Routing table is present in RSU's cache} \\ 0, \text{Otherwise} \end{cases} \quad (3)$$

When,
$\delta_{V,R} = 1$: RSU forwards the routing information to the assigned vehicle.
$\delta_{V,R} = 0$: RSU cache does not include routing information, hence initiating next stage of operation.

In the next stage, fittest vehicles from every hop are chosen for packet dissemination.

Phase 2

1. Transport network is presented as a directed graph, $G = (V, E)$
Where,
$V = [v_1, v_2, \ldots, v_N]$ - Collection of Vehicles
$E = [E_1, E_2, \ldots, E_N]$ - Links connecting the vehicles
Let, 'v_a' and 'v_b' be vehicles along the path with primary position vector (X_{a_0}, Y_{a_0}) and (X_{b_0}, Y_{b_0}) and initial speed (S_{a_0}, S_{b_0}). Vehicles which are neighbours are single hop away from one another. Eq. (4) represents the gap $(D_{i,j})$ existing between them with communication range $(C_{i,j})$ [28].
The link is found when $D_{i,j} < C$

$$D_{i,j} = \sqrt{(X_{a_0}, X_{b_0})^2 + (Y_{a_0}, Y_{b_0})^2} \quad (4)$$

2. In Eq. (5), the barking co-efficient (B_c^v) determines the fittest neighbour for transmitting packets. 'B_c^v' is highly dependent on Link Stability (L_s), Probability of effective message reception $\left(P_{i,j}^r\right)$ and Collision Probability (C_p). The next fittest neighbour for transmitting packets can be determined based on its value. '$V_{High}(B_c^v)$' represents the fittest vehicle.

$$B_c^v = \frac{L_s + P_{i,j}^r}{C_p} \quad (5)$$

3. Link between 'v_a' and 'v_b' is established at 't_0' when $D_{i,j} < C$, where '$D_{i,j}$' is the distance between them. Link is broken at 't_1' when $D_{i,j} > C$. Link duration is given by,

$$L_D = t_1 - t_0 \quad (6)$$

'L_s' from a hop to another [26] is given by,

$$L_s = \frac{L_D}{Max_{L_D}} \quad (7)$$

Where,
Max_{L_D} - Routing table's constant validity period

4. Let, '$n_{packets}$' be the quantity of packet transmissions in current hop (H_c) and 'n_{min}' be minimum number of packets transmitted to evade collision. 'C_p' is given by

$$C_p = 1 - \frac{n_{packets} - n_{min}}{n_{packets}} \tag{8}$$

Where,

$$n_{packets} = n - \varphi + \prod_{i=0}^{H_{adj}} \tau \tag{9}$$

Where,

n - Amount of vehicles in 'H_c'

φ - Amount of vehicles which stop transmission on obtaining control packet from current hop

The vehicular network involves multi-hop communication.

$\prod_{i=0}^{H_{adj}} \tau$ - Neighbouring hop packets

H_{adj} - Amount of interfering hops

5. '$P^r_{i,j}$' is determined depending on distance amid adjoining vehicles. Nakagami fading model [29] is used for capturing movement of vehicles in urban area [30]. 'P^r' is computed based on fading parameter (ψ),

$$P^r = e^{-\psi\left(\frac{D^2}{C}\right)} \sum_{i=1}^{\psi} \frac{\left(\psi\left(\frac{D^2}{C}\right)\right)^{i-1}}{(i-1)!} \tag{10}$$

6. Based on 'B_c^v', best candidate (vehicle) is chosen for sending packets. '$Pos_{High}(B_c^v)$' is best location where 'B_c^v' is maximum.

Where,

φ - Acceleration constant.

ran - Random number in range [0,1].

$V_i(B_c^v)$ - Velocity vector of 'i^{th}' vehicle.

$Pos_i(B_c^v)$ - Position vector of 'i^{th}' vehicle.

$V_{next}(B_c^v)$ - Velocity of ensuing vehicles.

$Pos_{next}(B_c^v)$ - Position of ensuing vehicles

$$V_{next}(B_c^v) = V_{Curr}(B_c^v) + \alpha \times ran\left(Pos_{High}(B_c^v) - Pos_{Curr}(B_c^v)\right) \tag{11}$$

$$Pos_{next} = Pos_{Curr} + V_{next}(B_c^v) \tag{12}$$

In Eq. (10), fitness function (f) is an aggregative reducing function with standardized routing measures.

W_1, W_2, W_3 and W_4 - Weight co-efficients.

PDR - Packet Delivery Ratio.

t_{Del} - Packet transmission delay.

Th - Throughput.

R - Routing overhead

$$f = W_1.(-PDR) + W_2.t_{Del} + W_3.R + W_4.(-Th) \tag{12'}$$

Subject to:

$$\sum_{i=1}^{4} W_i = 1, \ldots W_i \in (0,1) \tag{13}$$

$$dis(V_{next}, V_{Curr}) \leqslant D_{max} \tag{14}$$

Fitness function deals with maximising PDR as well as throughput and reducing overhead as well as delay.

In addition, Fig. 1 presents the detailed view of the proposed KMRFA used for determining optimal route in VANETs.

3.3. Komodo Mlipir Algorithm (KMA)

This Komodo Mlipir Algorithm (KMA) is proposed depending on the inspiration derived from Komodo dragons. The Komodo dragons are considered as ferocious reptiles which possesses the capability of searching food even when the distance between them and the prey is several kilometers away. This KMA algorithm simulates the reproduction, foraging and social hierarchy of the Komodo dragons. Depending on the fitness, the individuals of Komodo dragons are categorized into small males, females and large males. The female Komodo dragons are optimized depending on their pattern of reproduction, while the small and large males are optimized depending on the pattern of movement towards the preys. This KMA algorithm in specific included the concept of Mlipir which is the movement of small males from the large males since the large males has the habit occasionally preying on weaker small males (weak siblings) and larvae. This movement of Mlipir also represents the movement of small males that forages on leftovers from the large male population of Komodo dragons. But this movement also introduces the predation risks posed from the large male population such that they prevent the possibility of them to feed safely. In this KMA algorithm, the high fitness large males attracts the low-quality males. On the other hand, the low-quality makes are either rejected or attracted by high quality males with the probability of 0.5 for the purpose of determining a new position. Further, for a simple optimization problem with 2-D function,

$$f(x_1, x_2) = x_1^2 + x_2^2 \tag{15}$$

Where,

x_1, x_2 - Horizontal, Vertical axes.

Global optimal solution which is the target is '$f = 0$' at '$x_1 = x_2 = 0$'.

Consider an initial population comprising of 6 Komodo individuals ($K_1, K_2, \ldots K_6$) representing candidate solutions is generated at random, where every Komodo represents vector involving only position of a candidate solution. Population is divided into the following groups:

- Big Males (BMs) of high quality
- Females of middle quality
- Small Males (SMs) of low-quality

Here, BMs of high-quality mimic the robust Komodo which catches the prey representing best solution.

Evolution is carried out in a specific order. BMs interact with one other for performing High-Exploitation Low-Exploration (HILE). With probability of 0.5, females either mate with the best BM or reproduce through parthenogenesis by performing Medium-Exploitation Medium-Exploration (MIME). Lastly, SMs perform mlipir movement and gather around the search space to carry out Low-Exploitation High-Exploration (LIHE) Further two Komodo dragons 'K_1' and 'K_2' which are BMs fight with one another. It is adopted as attraction as well as distraction using particular rule. Low-quality BMs move forward to high-quality ones. A high-quality BM may move backward or forward to low-quality ones (probability=0.5). Some BMs perform intensification (high-exploitation) in a small area round about global optimum, whereas others perform diversification (low-exploration) in broader area.

Assume that 'K_1' attracts 'K_2' which makes it to move to global optimum (exploitation). This movement depicts the behaviour of Komodo to fight for prey with another BM. On the other hand, 'K_1' is distracted from 'K_2' at random (probability=0.5) and perform exploration. This movement mimics the behaviour of Komodo to retain the prey with itself. KMA is developed to preserve high-quality BMs. Females either mates with Winners (maximum fitness) to produce 2 offspring's or involve in parthenogenesis by moving randomly. Therefore, a middle-quality female hunts for solution in MIME manner. SMs of low quality, Komodo dragons (K_4, K_5 and K_6) perform a mlipir to move BMs by arbitrarily choosing a portion of dimensions. Dimensions are chosen by using normal distribution (probability=0.5). This kind of movement seems to be new in case of meta-heuristic optimization. The movement as well as probability of dimensional share are called 'mlipir' and 'mlipir rate' respectively. FThe mlipir movement involves LIHE. Hence, innovative movements play a dominant role. As population in early generations involve increased diversity, the mlipir movement makes SMs do

perform increased exploration covering the complete search space. They perform a LE during last iterations.

The movements of BMs, females and SMs may be applied to high-dimensional (m) spaces. BMs attract and distract one another in complete m-dimensions, while SMs perform a mlipir movement to BMs only in 'dm' dimensions, where 'd' represents mlipir rate denoted as real number within range (0, 1) to choose a portion of dimensions. Every new position of SMs is maintained without any survivor selection.

Algorithm. KMA.

Output: K_{Best} - Global optimal solution
Set 'n', 'p' and 'd'
Initialise 'n' individuals involving 'm' dimensions.
while ($Stopping_{Criterion} = 0$) do
    for (every Komodo)
        Determine quality and rank them.
        Split population into groups: BMs(b), middle-quality female (1) and SMs (s) using Eqs. (16) and (17) based on ranks as well as 'p'
        end
    for(every BM)
        Move using Eq. (19)
        Keep 'q' BMs (best locations) to persist in next generation.
        Modify female by mating winner BM by using Eq. (20) & Eq. (21) or performing parthenogenesis using Eq. (23)
    end
    for(every SM)
        Move using Eq. (25) and preserve new locations for ensuing generation.
        Modify 'n' using Eq. (26)
        Choose Komodo of high quality as best solution (K_{Best})
    end
end

First, quantity of Komodo (n), portion of BMs (p) along with mlipir rate (d) are set. 'n' Komodo are created at random. Evolution takes place until the condition is fulfilled. In every generation, Komodo are assessed depending on problem's objective. Qualities are organised to define ranks using which population is divided into BMs(b), middle-quality female (1) and SMs (s) depending on Eqs. (16) and (17). Then Every BM moves as shown in Eq. (19), and the 'b' best locations are kept for surviving to ensuing generation. Then, females either mate or carry out parthenogenesis as shown in Eq. (20) & Eq. (21), and Eq. (22) respectively. Every SM moves as shown in Eq. (25), and their new locations survive in ensuing generation (without survivor selection). Population adaptively updates by using Eq. (26). Finally, high quality Komodo (K_{best}) is returned as global optimal solution.

KMA involves 2 phases to effectively handle unimodal as well as multimodal functions. Initially, KMA employs small population involving half portion of BMs and high mlipir rate. The stage is run for around 1000 generations for determining problem complexity. In case the problem is unimodal involving less complexity, KMA determines global optimum quickly. On reaching the global optimum, solution is returned, and evolution is stopped. Else, it moves to the next stage and executes it until maximum function assessments are done or algorithm attains global optimum. KMA uses better population which is capable of self-adapting to the size of population during evolution with half number of BMs and 'd'. The algorithm is found to solve multimodal (complex) benchmarks.

Representation of Individual.

Every individual is represented as real-valued vector 'K' of size 'm'. It finds a location in problem space. Like fireflies in FA, every Komodo in KMA has only position. As inertia is not considered during movement, it moves dynamically.

Groups of individuals.

Population is divided into following groups: BMs of high-quality, females with middle-quality, and SMs of low-quality with a share 'p' that is in interval (0, 1) and may be adjusted to 0.5 for a problem. By using 'p', Komodo (n) is divided into BMs(q), middle-quality female (1) and SMs (s) using Eqs. (16) and (17).

$$b = \lfloor (p-1).n \rfloor \tag{16}$$

$$s = n - b \tag{17}$$

Nevertheless, a highly small or big portion (p) produces '0' for 's' or 'b'. A simple method is followed to apply BMs and SMs with at least 2 Komodo communications.

Movements of BMs.

BMs interact through attraction/ distraction by using a simple rule. A BM of low-quality is attracted by high-quality one. On the other hand, a BM of high-quality may be attracted/ distracted by ones with low quality at random (probability=0.5) to obtain a fresh location. This mechanism

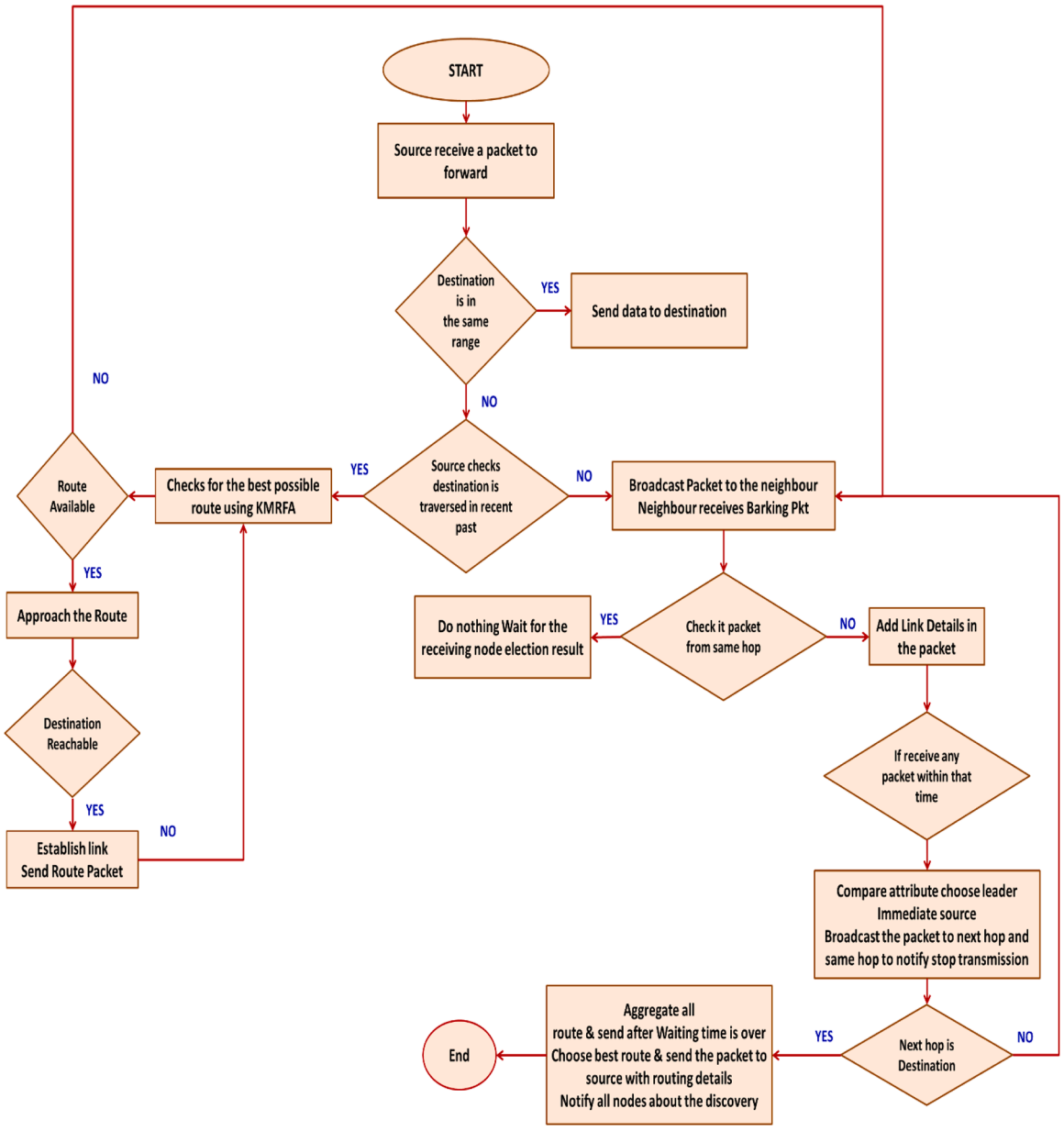


Fig. 1. Detailed view of the proposed KMRFA used for determining optimal route.

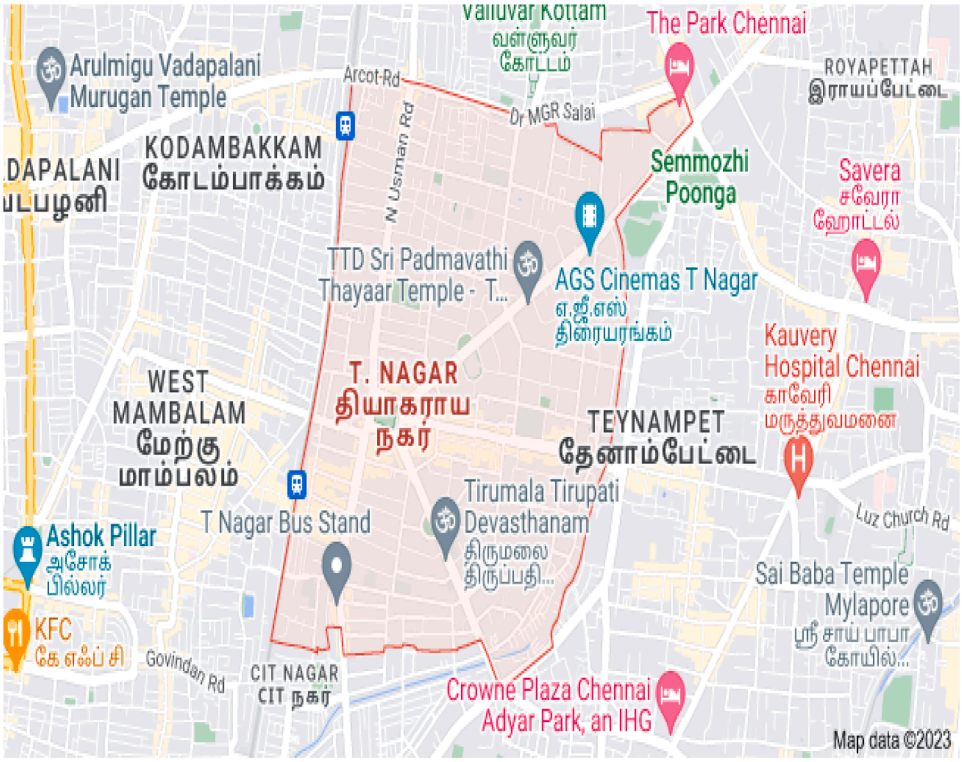


Fig. 2. Open Street map of the simulation area which is imported into SUMO.

guarantees that chance of exploitation is more in contrast to exploration. Hence, this movement mechanism is called HILE.

BM's (K_i) movement to generate a location '$K_i^{'}$' is defined as shown below.

$$\omega_{ij} = \begin{cases} r_1\left(K_j - K_i\right), \text{if } f\left(K_j\right) < f(K_i) \text{or} r_2 < 0.5 \\ r_1.\left(K_i - K_j\right), \text{Otherwise} \end{cases} \tag{18}$$

$$K_i^{'} = K_i + \sum_{j=1}^{b} \omega_{ij}, j \neq i \tag{19}$$

Where,

$f(K_i), f\left(K_j\right)$ - Fitness of 'i^{th}' and 'j^{th}' BMs.

K_i, K_j - 'i^{th}' and 'j^{th}' BMs.

r_1, r_2 - Random numbers in range [0,1].

q - Quantity of BMs.

In case there are 2 BMs, then the one with low-quality does the exploitation, while the one with high-quality does exploration or exploitation at random (probability=0.5). BMs exhibit 2 types of motion.

- **Movement 1:** BMs perform exploitative search in regions near global solution.
- **Movement 2:** Low-quality BM performs exploitation by attracting high-quality ones, while high-quality BM does exploration by distracting low-quality ones. Hence, the likelihood of exploitation is more than exploration.

If population includes 3 BMs, then 8 movements of BMs are possible.

- **Movement 1:** All BMs perform exploitation close to global optimum.
- **Movements 2 - 5:** 2 BMs perform exploitation, and 1 does exploration.
- **Movements 6 - 8:** 1 BM performs exploitation, while other 2 do exploration.

In all movements, at least 1 BM is involved in exploitation, and there is no likelihood of all 3 BMs performing exploration. Hence, it is assured that possibility of exploitation is more than exploration.

Nevertheless, an optimum quantity of BMs is essential to carry out interaction. Theoretically, 2 or 3 BMs offer optimal interaction to attain a balance between exploitation and exploration. 2 BMs are sufficient to deal with unimodal functions. However, to handle multimodal functions, 3 BMs are essential.

Female Reproduction.

A female of middle-quality is expected to perform exploitation/exploration with a probability of 0.5. If exploitation is chosen, females strengthen the region around global optimal solution by mating with winner BM to produce 2 offspring's (new locations). Else, it varies solution in a wider search space by performing parthenogenesis. Finally, females are replaced by best-produced offspring. Further, mating method or sexual reproduction of 2 Komodo is given below.

$$K_{il}^{'} = \dot{r}_l.K_{il} + (1 - \dot{r}_l\ ).K_{jl} \tag{20}$$

$$K_{jl}^{'} = \dot{r}_l.K_{jl} + (1 - \dot{r}_l\ ).K_{Il} \tag{21}$$

where,

K_{il}, K_{jl} - 'l^{th}' dimension of 'i^{th}' and 'j^{th}' Komodo as winner BM and female.

$K_{il}^{'}$, $K_{jl}^{'}$ - 'k^{th}' dimension of 2 offsprings generated by mating.

$\dot{r}_k$ - Random number in range [0,1] for 'l^{th}' dimension.

Offsprings are produced as mating is randomly performed in every dimension. Two offspring's are produced and the female is replaced by best offspring.

Parthenogenesis is implemented by adding a small number which is produced at random using symmetric normal distribution to every female dimension. The number is given by,

$$\left(K_{i1}^{'}, K_{i2}^{'}, \ldots K_{im}^{'}\right) = \left(K_{i1}, K_{i2}, \ldots, K_{im}\right) \tag{22}$$

$$K_{ij}^{'} = K_{ij} + (2r-1).\varphi.\left(U_j - L_j\right) \quad (23)$$

Where,

$K_{i1}, K_{i2}, \ldots, K_{im} \in [L_j, U_j]$ - Vector elements of 'm' dimensions of 'K'.

L_j, U_j - Lower and Upper bounds of 'j^{th}' dimension.

r - Random number in normal distribution.

φ - Parthenogenesis radius (set to 0.1).

'$\varphi = 0.1$' means that solution may be produced in a radius of 10 % of whole search space.

Movements of SMs.

A SM Komodo has 2 goals:

- Search for leftovers of BMs
- Overcome cannibalism by moving away (mlipir)

This behavior is implemented in the algorithm through mlipir movement. SMs of low-quality are involved in solution search in a broader region, but sometimes perform less amount of exploitation. Every SM moves away from BMs that is applied by arbitrarily choosing a portion of dimensions with specific probability (d). With 'd' in the range $[0, 1]$, mlipir movement of 'i^{th}' Komodo that follows 'j^{th}' one is given by,

$$\omega_{ij} = \begin{cases} \sum_{l=1}^{m} r_1\left(K_{jl} - K_{il}\right), \text{if} r_2 < d \\ 0, \text{Otherwise} \end{cases} \quad (24)$$

$$K_i^{'} = K_i + \sum_{j=1}^{q} \omega_{ij}, \text{where } j \neq 1 \quad (25)$$

Where,

r_1, r_2 - Random numbers in range [0,1].

'r_1' defines movement velocity.

'r_2' chooses dimension to follow.

m - Dimension.

K_{il}, K_{jl} - 'l^{th}' dimension of 'i^{th}' SM and 'j^{th}' BM.

l - Randomly chosen dimension depending on normal distribution with probability of 'd' to choose $1, 2, .., m-1$ dimensions of BM.

b - Quantity of BMs.

For 2 SMs and 2 BMs, 9 possible movement combinations of SMs are possible.

- **Movements 1 & 2, 4 & 5:** Both SMs perform explorative search by moving horizontally as well as vertically to gather around BMs in all directions.
- **Movements 3, 6 - 9:** At least 1 SM performs exploitation by moving along the diagonal nearer to BM

If movements of SMs are computed separately, then 12/18 movements are explorations, while rest are exploitations. In mlipir movement, the likelihood of exploration is assured to be better in contrast to exploitation.

Population Adaptation Mechanism.

The parameter 'n' is highly sensitive when compared to 'p' and 'd' as it has an intense control on exploration as well as exploitation approaches. Parameters 'p' and 'd' are assigned a fixed value of 0.5 as problem features are not known. Hence, an adaptation mechanism is proposed for adaptively tuning 'n' during evolution. If there are 2 successive fitness enhancements, 'n' is reduced by removing 5 individuals. Else if stagnation is observed, 'n' increases as 5 fresh individuals are produced. Fresh individual is produced from best BM which is randomly moved. The size of population ($n^{'}$) is computed as shown below:

$$n^{'} = \begin{cases} n - n_i, \text{if} \delta f_1 > 0, \delta f_2 > 0 \\ n + n_i, \text{if} \delta f_1 = 0, \delta f_2 = 0 \end{cases} \quad (26)$$

Where,

n_i - Number of individuals to remove or produce (set to 5)

$$\delta f_1 = \frac{|f_1 - f_2|}{f_1}$$

$$\delta f_2 = \frac{|f_2 - f_3|}{f_2}$$

$\delta f_1, \delta f_2$ - Fitness difference of 'i^{th}' and '$i-1^{th}$' generations.

'n_i' signifies a small population employed in initial stage. Minimum (20), Initial (200) and maximum (200) values of 'n' are to be subsequently set.

3.4. Complexity analysis of the proposed KMAORDM scheme

The complexity analysis of the proposed KMAORDM scheme is determined based on the three operations that includes vehicular nodes initialization, the fitness function used for assessing the potentiality of each vehicle and the process of CHs updating. The first operation of vehicular nodes initialization during the implementation of the proposed KMAORDM scheme is $O(N_V)$ for N_V number of vehicles. Then the location vectors related to all vehicles are updated for determining the best position. The computational complexity related to the process of position updating is $O(Max_{Iter}N_V) + O(Max_{Iter}N_V P_{Dim})$. Where, Max_{Iter} and P_{Dim} represents the maximum number of iterations and problem dimension, respectively. Hence the computational complexity of the proposed KMAORDM scheme is $O(N_V(Max_{Iter} + Max_{Iter}P_{Dim} + 1)$.

4. Results and discussion

The performance evaluation of the proposed KMRFA algorithm and the benchmarked CORFA [25], GWO-MHRLM [24] and LBAFP [23] approaches are conducted in terms of QoS factors using the system implemented using the open-source traffic simulator named SUMO. These benchmarked approaches are considered for evaluation since they are recently proposed bio-inspired optimal route determination approaches contributed for enhancing the degree of QoS in VANETs. The simulation environment considered for the proposed KMAORDM method and the baseline CORFA, GWO-MHRLM and LBAFP approaches are the same for preventing the bias during the process of implementation. The results of these approaches are averaged over 30 runs, and the results are plotted for determining the optimal among the them in satisfying the requirements of QoS under the process of data routing.

Initially, the Open Steer Map of Thiagarajah Nagar in Chennai is imported as xml,format and named as osm,xml. This file comprises of complete lanes, streets and buildings which aids in constructing a real time platform of simulation. Then the osm,xml is converted into net.xml file using Netconvert. Further ActivityGen package is used for generated vehicle mobility during the phase of network initialization. This ActivityGen package helps in creating the number of source and destination pair for each vehicle that is saved in trips.xml. Finally, DUAROUTER package is used for generating route file rou.xml. MATLAB is used for implementing the vehicular nodes communication depending on the proposed routing mechanism. Moreover, traffic control interface (TRACI) being a TCP-based sever client architecture is used for initiating the interface between MATLAB and SUMO. In addition, Fig. 2 presents the OSM map of Thiagarajah Nagar in Chennai.

The simulation environment comprises of 500 vehicular nodes which is distributed over the network area of 2000 m × 2000 m with 250 m and 150 m is set as the range of communication with respect to the modes of V2I and V2V, respectively. The size of population (Komodo dragon) used for implementation is 100 with the maximum number of iterations set to 100. The velocity of the vehicle in the two-dimensional network search plane is 22–30 m per second. The number of simulation runs is 10 with the convergence factor set to 0.001. Moreover, the mobility model considered for implementation is the Freeway mobility

Table 1
Summary of the existing bio-inspired routing protocols contributed to the literature for VANETs scenario.

Authors	Mechanism Used	Merits	Limitations
Rajput et al.[16]	Hybrid ACO, PSO, Cuckoo and Firefly swarm intelligent routing algorithm (HSIRA)	It was proposed for determining optimal solution with the modification of ACO, PSO, Cuckoo Search and Firefly algorithms by solving multi-constraint QoS optimal path limitation using a potentially developed cost function	The establishment of link maintenance between vehicular nodes is highly challenging in dynamic mobile vehicular nodes
Ramasamy et al. [17]	Swarm intelligent routing algorithm using Fuzzy-based PSO (FPSOA)	This FPSOA algorithm provisioned a high-quality route for communication between the vehicular nodes through the inclusion of two steps such as selection for candidate nodes and construction of optimized routing model	The determination of shortest path is not always the single objective in dynamic and scalable VANET
Ramamoorthy and Thangavelu[18]	Hybrid Ant Colony Optimization Routing Protocol (EHACORP)	This protocol determined the shortest path depending on the factors that included minimized number of hops between source and destination nodes and reduced communication costs.	The vehicular environment considered during implementations is non-realistic
Attia et al.[19]	An Advanced Greedy Hybrid Bio-Inspired (AGHBI) routing protocol	This AGHBL routing mechanism was developed a greedy approach which concentrated towards the improvement of network performance by specifically utilized the benefits of bee colony optimization for maintaining the path with reduced workflow and selecting the optimized route with maximized QoS.	Communication overhead during route discovery and maintenance was comparably greater. In addition, the process of Communication between zones is also challenging
Kumbhar et al.[20]	UniFaVa Routing protocol	This UniFaVa used vehicular nodes with sufficient residual energy as the relay and analytically predicted the time of connectivity for ensuring reliable data transmission during V2V mode of communication	It did not consider alternate routes and link stability into account during the process of route determination.
Ramamoorthy and Thangavelu[21]	Energy-Efficient Shortest Path Without Congestion (EESPWC) algorithm	This EESPWC algorithm was proposed as a reliable routing algorithm that targeted on the satisfaction of the parameters that included reduced energy consumption, low packet loss ratio, low end-to-end delay, low routing overhead and increase throughput.	It ignored the significance of cache-assisted RSUs during the process of data routing.
Ramamoorthy and Thangavelu[22]	Enhanced ACO (EACO)-based Shortest Route Determination algorithm	It mainly concentrated on the links that possesses minimum hops through the identification of high level of connectivity and selecting path with shortest distance established between the vehicular nodes.	The computation and communication overhead realized in the network is comparatively high under urban VANETs.
Seth et al.[23]	Lane-based advanced forwarding protocol (LBAFP)	This LBAFP protocol uses the vehicular nodes' position coordinates for identifying the destination zone for facilitating packets' broadcasting towards the destination direction. It also used the broadcasting method of suppression for minimizing the network routing overhead such that interferences that arises with respect to various road segments are completely minimized.	Only limited number of lanes are monitored during routing process which may not be ideal for real time traffic scenario.
Nahar and Das[24]	Meta-Heuristic GWO-based Reinforcement Learning method (GWO-MHRLM)	It possessed the merits of global search which in turn inherited parameterized strategy which helped in eliminating uncertainty in rewards and exploration of vehicular nodes' state that helps in optimizing the structure of multilevel network.	It failed in handling the change in the network traffic concerned with urban VANET scenario.
Ghosh et al.[25]	Canine Olfactory Route-Finding Algorithm (CORFA)	It adopted the canine's exceptional potential for acquiring knowledge about the environment such that message can be effectively exchanged between the neighbouring nodes.	The factors considered for determining optimal routes between source and destination vehicles is not comprehensive.

model. Further the simulation experiments are conducted with Nakagami propagation model and IEEE 802.11p as the physical layer radio model. In addition, the simulation parameters used for implementing the proposed KMRFA algorithm and the benchmarked approaches is listed in Table 2.

4.1. Performance evaluation based on based on different vehicular density

In this section, the results of the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches evaluated using Throughput, Packet Delivery Rate, Routing Overhead and Network Latency with different vehicular density are presented. In specific, Figs. 3 and 4 demonstrates the plots of Throughput and Packet Delivery Rate achieved by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different vehicular density. It is visualized that the number of hops determined for establishing the routing between the source and destination during the implementation of the proposed KMAORDM is significantly reduced. It also adopted the fitness function which included the factors of energy, distance, number of hops and link stability into account for establishing routing such that delivery of packets is predominantly improved independent to the number of data packets generated by the increase in the number of vehicular nodes. Thus, the proposed KMAORDM under different vehicular density confirmed maximized throughput of 14.98 %, 17.19 % and 21.42 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the packet delivery rate achieved by the proposed KMAORDM under different vehicular density is improved by 15.68 %, 18.76 % and 21.18 %, better than the benchmarked approaches.

Moreover, Figs. 5 and 6 unveils the plots of routing overhead and network latency incurred by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different vehicular density. The proposed KMAORDM adopted the significant movement of mlipir to explore the route between the source and destination vehicular nodes such that delay of packets reaching the required target is considerably minimized in the network. It also adopted a phenomenal merit of RSUs cache into the routing approach such that history of data routing utilized in previous iterations is available depending on the requirement of routing process. Hence the proposed KMAORDM under different vehicular density guaranteed reduced routing overhead of 17.12 %, 21.98 % and 23.26 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the network latency incurred by the proposed KMAORDM under different vehicular density is improved by 16.18 %, 19.14 % and 22.64 %, better than the benchmarked approaches.

Table 2
Simulation parameters used for implementing KMRFA algorithm and the benchmarked approaches.

Simulation Parameters	Values used
Number of vehicles	500
Network Area	2000 m × 2000 m
Simulation time	3600 s
Traffic Type	CBR
Data packet size	2816 bytes
Range of Communication (V2I)	300 m
Range of Communication (V2V)	150 m
Propagation model	Nakagami
Carrier frequency	5.89 GHz
MAC type	IEEE 802.11p

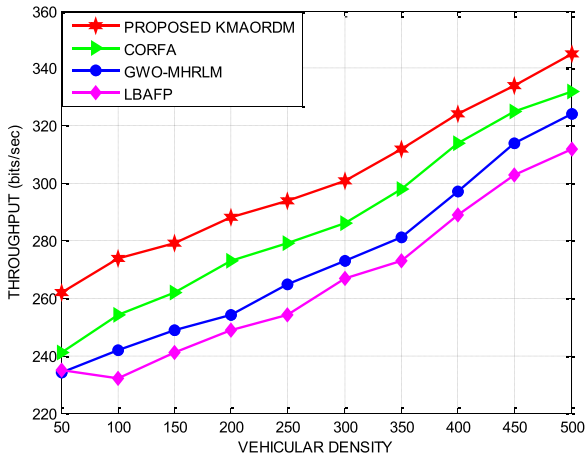


Fig. 3. Proposed KMOAEDM-Throughput with different vehicular nodes.

4.2. Performance evaluation based on based on maximum speed of vehicles

In this section, the results of the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches evaluated using Throughput, Packet Delivery Rate, Routing Overhead and Network Latency with different maximum speed of vehicles are presented. In specific, Figs. 7 and 8 demonstrates the plots of Throughput and Packet Delivery Rate achieved by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different

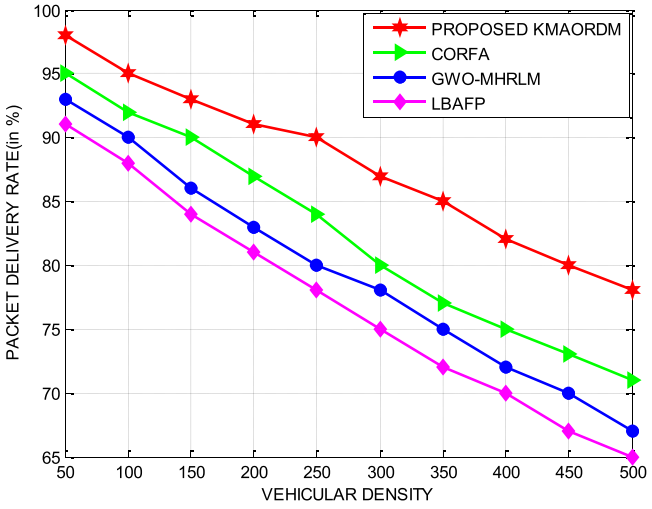


Fig. 4. Proposed KMOAEDM-Packet Delivery Rate with different vehicular nodes.

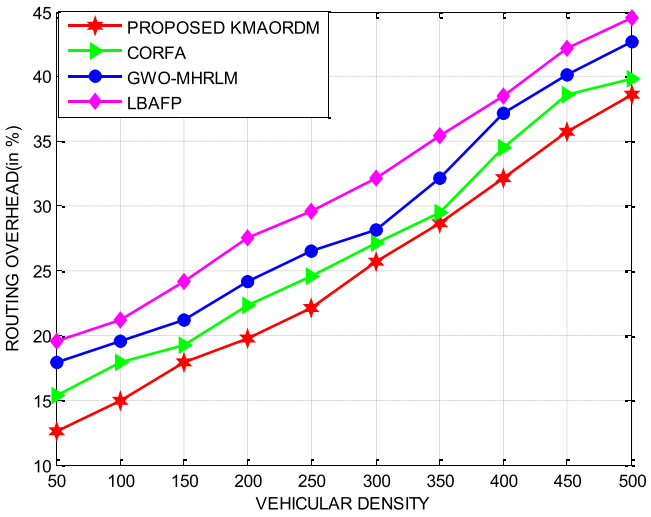


Fig. 5. Proposed KMOAEDM-Routing Overhead with different vehicular nodes.

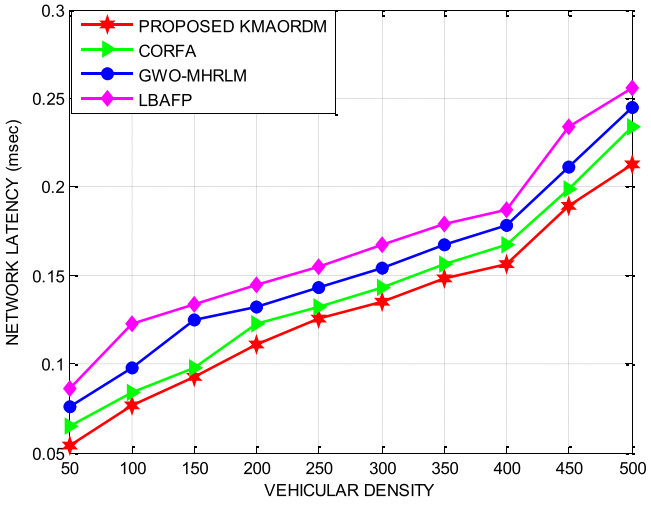


Fig. 6. Proposed KMOAEDM-Network latency with different vehicular nodes.

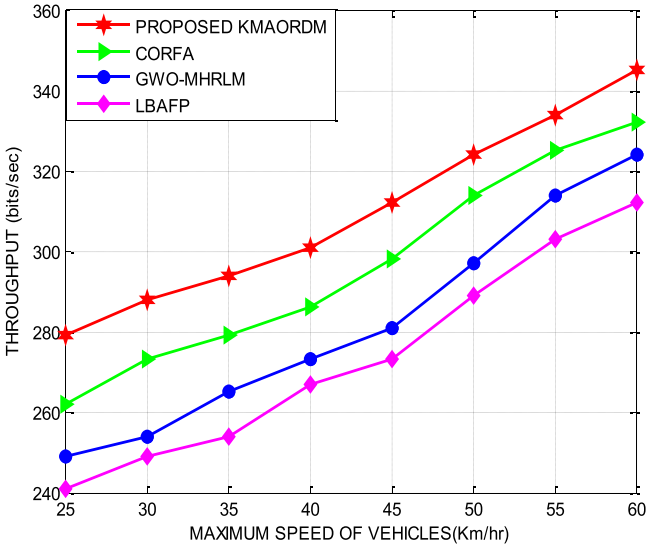


Fig. 7. Proposed KMOAEDM-Throughput with different maximum speed of vehicles.

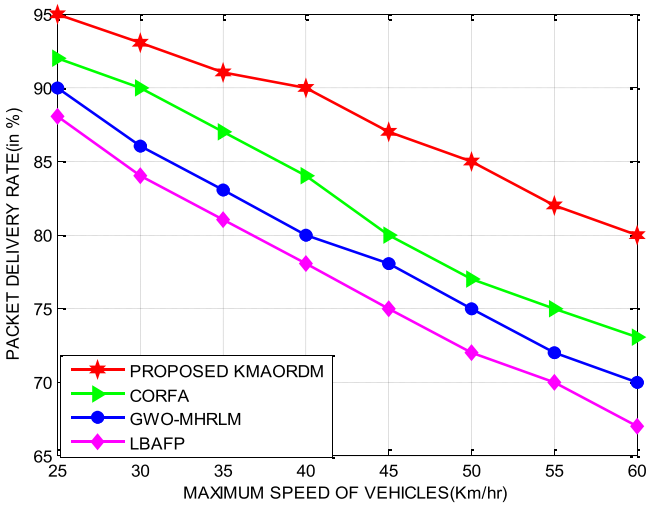


Fig. 8. Proposed KMOAEDM-Packet Delivery Rate with different maximum speed of vehicles.

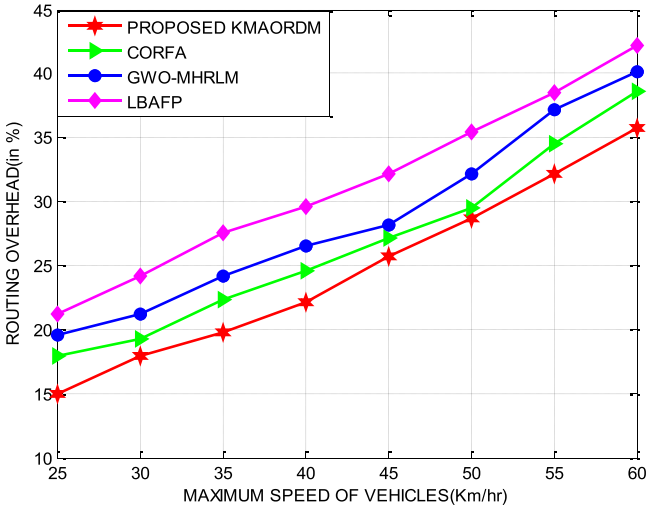


Fig. 9. Proposed KMOAEDM-Routing Overhead with different maximum speed of vehicles.

maximum speed of vehicles. It is observed that the proposed KMAORDM depending on the systematic increase in the maximum speed of vehicles incorporated a exploitation policy which determined the genuineness of vehicular nodes during the routing task. This adopted policy helped in finding the minimum hop route for data routing without intentional packet drop and retransmissions in the network. Thus, the proposed KMAORDM under different maximum speed of vehicles guaranteed maximized throughput of 14.69 %, 17.68 % and 20.14 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the packet delivery rate achieved by the proposed KMAORDM under different maximum speed of vehicles is improved by 15.56 %, 18.24 % and 20.12 %, better than the benchmarked approaches.

Moreover, Figs. 9 and 10 depicts the plots of routing overhead and network latency incurred by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different data packet size. It is determined that the proposed KMAORDM scheme balanced the factors of route determination in a more significant manner such that time incurred for routing is highly reduced independent to the maximum speed of vehicles. It is also realized that number of packets retransmission under low link stability is remarkably minimized since it always forwarded packets from the source to the destination through the trustworthy vehicular nodes in the network. It also included a mobility

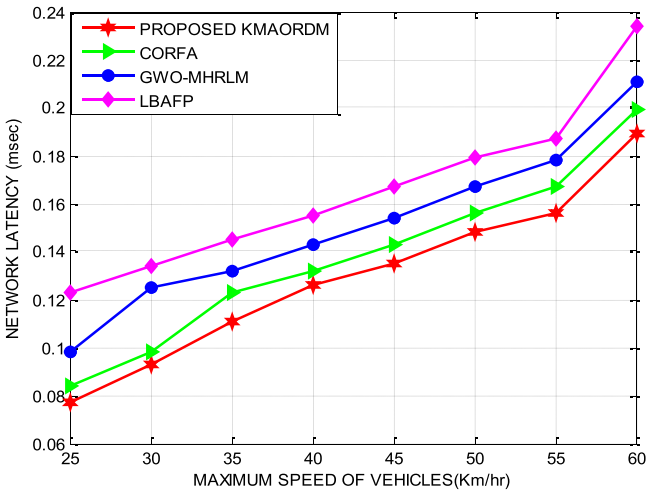


Fig. 10. Proposed KMOAEDM-Network latency with different maximum speed of vehicles.

factor which contextually helped in reducing the routing overhead to the expected level. Hence the proposed KMAORDM under different data packet size confirmed minimized routing overhead of 15.24 %, 18.79 % and 21.58 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the network latency incurred by the proposed KMAORDM under different data packet size is improved by 14.86 %, 17.28 % and 20.74 %, better than the benchmarked approaches.

4.3. Performance evaluation based on distance between source and destination vehicular nodes

In this section, the results of the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches evaluated using Throughput, Packet Delivery Rate, Routing Overhead and Network Latency with different distance between source and destination vehicular nodes. In specific, Figs. 11 and 12 demonstrates the plots of Throughput and Packet Delivery Rate achieved by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different data packet size.

With increase in the distance between the source and destination vehicular nodes, the proposed KMAORDM mechanism adopted a reactive strategy which explored possible dimensions of vehicular nodes which could contribute towards successful delivery of data packets. Thus, the proposed KMAORDM under different distance between source and destination vehicular nodes confirmed maximized throughput of 13.98 %, 16.32 % and 19.56 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the packet delivery rate achieved by the proposed KMAORDM under different distance between source and destination vehicular nodes is improved by 16.64 %, 20.91 % and 22.68 %, better than the benchmarked approaches.

Moreover, Figs. 13 and 14 unveils the plots of routing overhead and network latency incurred by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different distance between source and destination vehicular nodes. It is confirmed that the incorporation of KMRFA approach. Hence the proposed KMAORDM under different data packet size confirmed minimized routing overhead of 16.42 %, 18.96 % and 21.18 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the network latency incurred by the proposed KMAORDM under different data packet size is improved by 15.26 %, 18.49 % and 21.52 %, better than the benchmarked approaches.

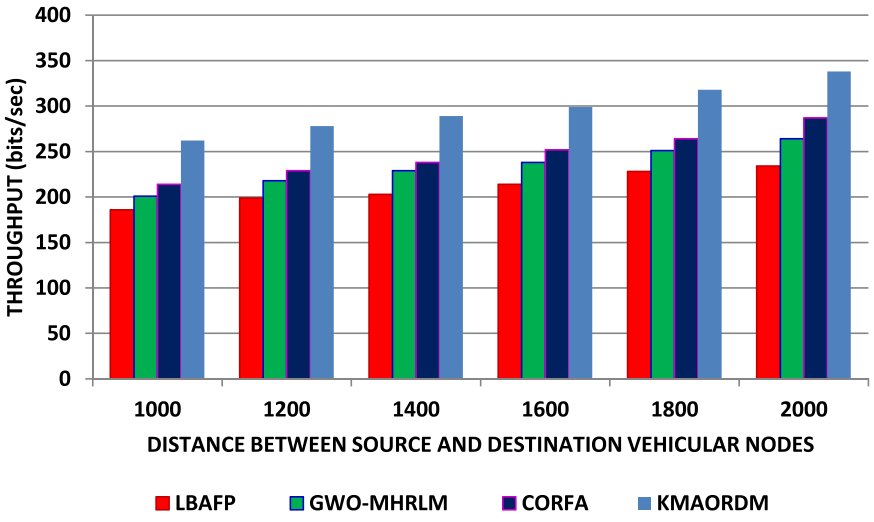


Fig. 11. Proposed KMOAEDM-Throughput with different distance between source and destination vehicular nodes.

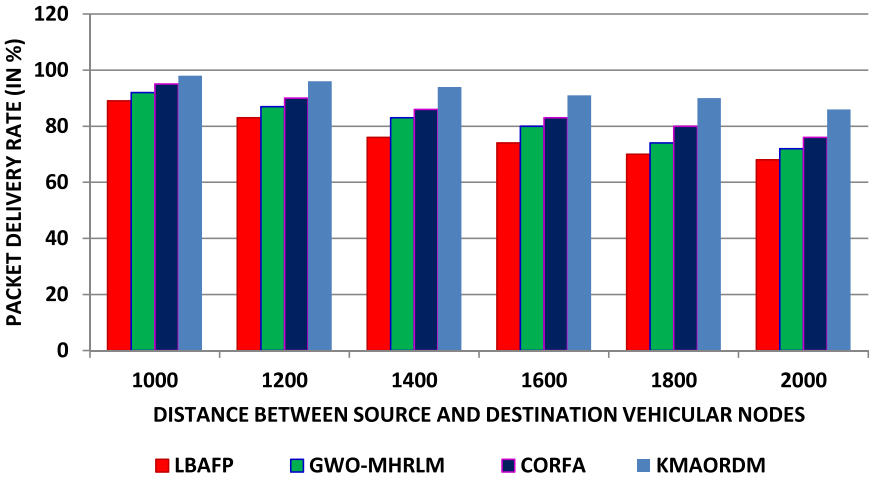


Fig. 12. Proposed KMOAEDM-Packet Delivery Rate with different distance between source and destination vehicular nodes.

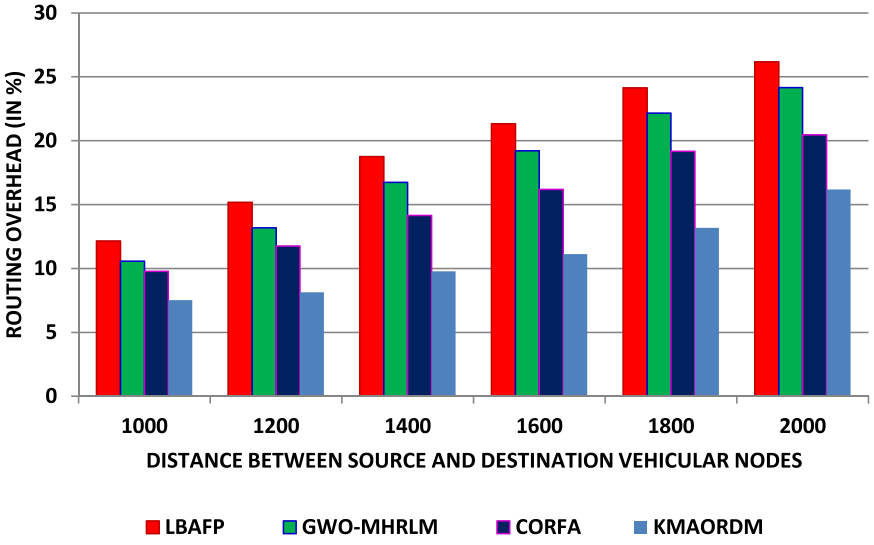


Fig. 13. Proposed KMOAEDM-Routing Overhead with different distance between source and destination vehicular nodes.

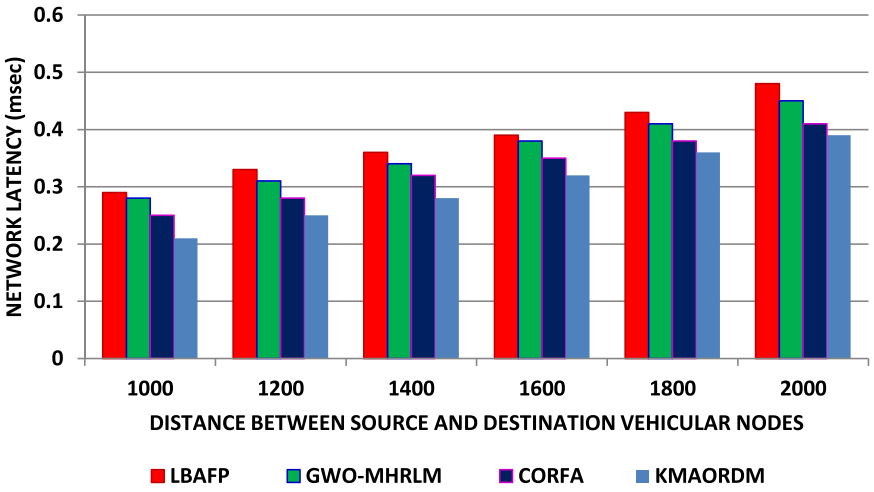


Fig. 14. Proposed KMOAEDM-Network Latency with different distance between source and destination vehicular nodes.

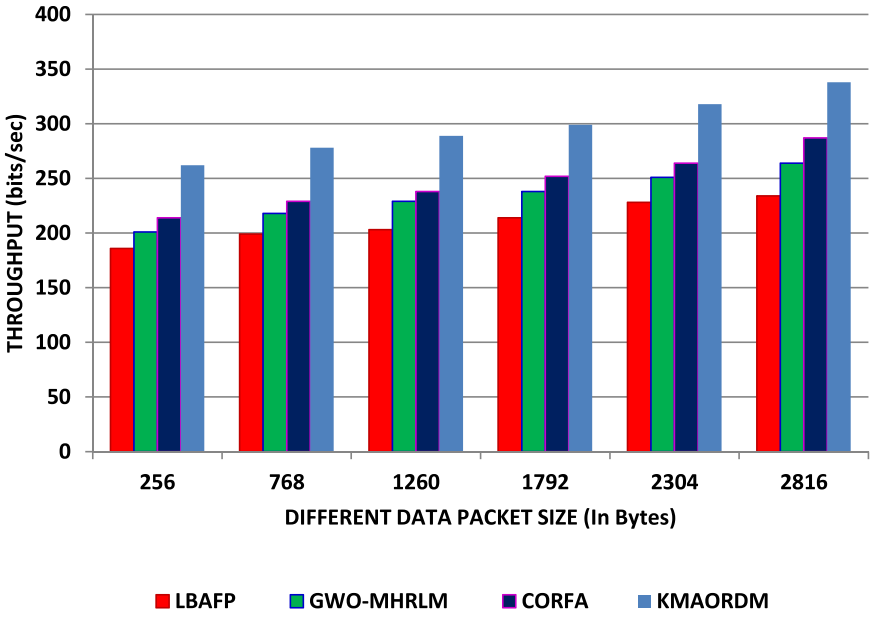


Fig. 15. Proposed KMOAEDM-Throughput with different data packet size.

4.4. Performance evaluation based on different data packet size in bytes

In this section, the results of the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches evaluated using Throughput, Packet Delivery Rate, Routing Overhead and Network Latency with different data packet size (in bytes) are presented. In specific, Figs. 15 and 16 presents the plots of Throughput and Packet Delivery Rate achieved by the proposed KMAORDM and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different data packet size. It is realized that the proposed KMAORDM is capable in balancing the trade-off between the rate of exploration and exploitation such that optimal route between the source and destination nodes is determined with maximized reliability. It was identified to adaptively handle the change in size of the data packets in a more potential manner, such that retransmission of packets is considerably minimized resulting in ideal delivery of packets to destination. Thus, the proposed KMAORDM under different data packet size confirmed maximized throughput of 15.68 %, 18.94 % and 22.19 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the packet delivery rate achieved by the proposed KMAORDM is improved by 17.32 %, 20.65 % and 23.49 %, better than the benchmarked approaches.

In addition, Figs. 17 and 18 depicts the plots of routing overhead and network latency incurred by the proposed KMAORDM scheme and the baseline CORFA, GWO-MHRLM and LBAFP approaches under different data packet size. It is identified that the adoption of KMRFA approach included three different phases that attributes towards better achievement of solution diversity in the network. This achievement of solution diversity only employed the routing through the determined reliable path which prevented unnecessary packet drop and delay in the delay. Hence the proposed KMAORDM under different data packet size confirmed minimized routing overhead of 16.94 %, 19.16 % and 22.32 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the network latency incurred by the proposed KMAORDM under different data packet size is improved by 15.42 %, 18.54 % and 21.16 %, better than the benchmarked approaches.

4.5. Statistical investigation

In this section, the statistical investigation of the proposed KMAORDM scheme and the baseline approaches is presented with respect to the tests of Fully Modified Least Square (FMOLS). This FMOLS tests include p-test, ANOVA, Regression analysis for exploring the

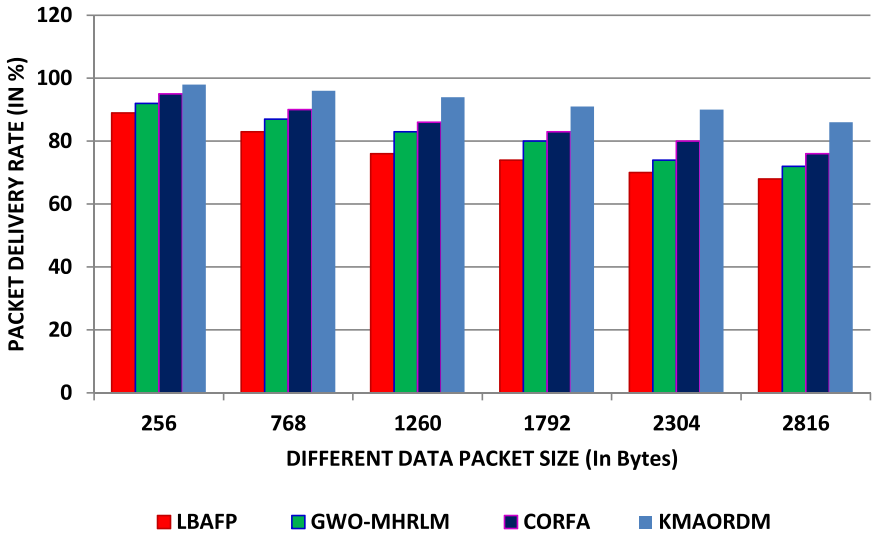


Fig. 16. Proposed KMOAEDM-Packet Delivery Rate with different data packet size.

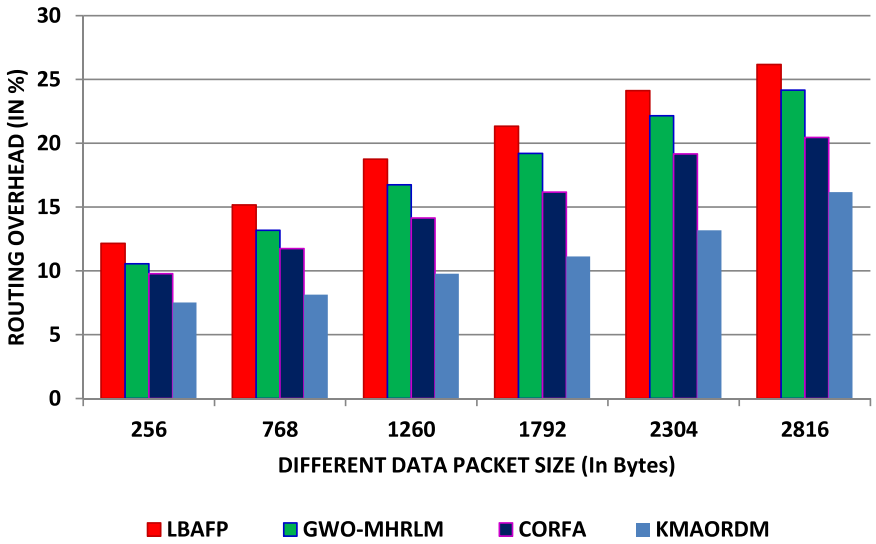


Fig. 17. Proposed KMOAEDM-Routing Overhead with different data packet size.

potential of the proposed KMAORDM scheme. In addition, Table 3 presents the results of the statistical investigation depending on regression analysis with respect to the number of clusters and the range of communication.

From the results, the impact of transmission range over the number of clusters constructed is determined. The fewer number of clusters are constructed with smaller range of communication. It is identified that the proposed KMAORDM scheme minimized the number of clusters by 0.44 % with increase in the range of transmission by 1.5 %. When the change in the transmission range, the baseline CORFA, GWO-MHRLM and LBAFP approaches such as is reduced by 0.041, 0.043 and 0.042, respectively.

4.6. Discussions

The significance of the improvements brought about by the proposed KMAORDM scheme and the benchmarked approaches as visualized from the simulation results is presented as follows.

i) The throughput and PDR confirmed by the proposed KMAORDM scheme with different densities of vehicular nodes is identified to be comparatively better than the baseline approaches since the number of clusters constructed in the network is significantly low which results in high stable network.
ii) With different vehicular densities, the proposed KMAORDM scheme due to the adoption of dynamic cache strategy minimized the network latency and routing overhead which prevents the possibility of huge packet retransmissions that are feasible in packet dropping scenarios.
iii) This proposed KMAORDM scheme independent of the speed of vehicles performed well in terms of throughput, packet delivery rate, routing overhead and network latency since it included the well-balanced rate of exploration and exploitation in the search space such that pre-mature convergence is prevented in the network.
iv) It is realized that the proposed KMAORDM is capable in achieving better solution diversity such that optimal route between the source and destination nodes is determined with maximized reliability is constructed even the size of data packets is increased in the network.
v) It was identified to include a dynamic strategy which adaptively handled the change in size of the data packets in a more potential manner, such that retransmission of packets is considerably minimized resulting in ideal delivery of packets to destination

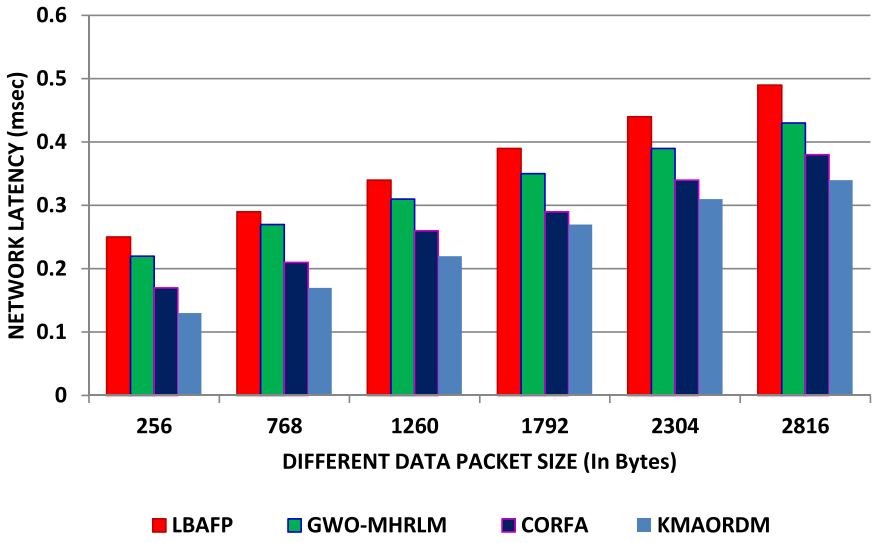


Fig. 18. Proposed KMOAEDM-Network Latency with different data packet size.

Table 3
Statistical Investigation Results of the proposed MAORDM scheme and the baseline approaches with respect to FMOLS.

Dependent Variables	Used Variable	Coefficient	Probability	R-Squared	Adjusted R-Squared	ANOVA
Number of clusters (NC) with LBAFP	RT	–0.053299	0.0099	0.765218	0.759884	F(1,9) = 41.92 ***
	NC	26.84306	0.0006			
Number of clusters (NC) with GWO-MHRLM	RT	–0.039189	0.0135	0.827314	0.785282	F(1,9) = 53.98 ***
	NC	28.12284	0.008			
Number of clusters (NC) with CORFA	RT	–0.041164	0.0098	0.889842	0.845318	F(1,9) = 68.14 ***
	NC	28.12741	0.0005			
Number of clusters (NC) with KMAORDM	RT	–0.042318	0.0121	0.912341	0.895212	F(1,9) = 74.56 ***
	NC	29.3218	0.0004			

5. Conclusion

The proposed KMRFA determined the route between the source and destination vehicles in VANETs which is an integral part of ITS. This KMRFA algorithm is a bio or nature-inspired approach which mimics the exceptional capability of Komodo Mlipir in terms of foraging and reproduction. This KMRFA algorithm is developed based on the inspiration of the populations' self-adaptation which helped in better control of balance between the degree of exploitation and exploration. This algorithm comprises of three main phases in which the first phase, better solution is determined through low exploration and high exploitation. In the second phase, the algorithm determines better solution either by using exploration or exploitation. This algorithm in the third phase diversifies the candidate solutions through the Mlipir movement for attaining maximized global solutions in the search space. This KMRFA algorithm determines the optimal route and the high fitness member used for packet forwarding through the inclusion of Mlipir coefficient. The performance of this KMRFA algorithm is evaluated using packet delivery rate, network latency, throughput and communication overhead with different vehicular density and speed of vehicles. The simulation results of the proposed KMAORDM under different vehicular density confirmed maximized throughput of 14.98 %, 17.19 % and 21.42 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. The packet delivery rate achieved by the proposed KMAORDM under different vehicular density is improved by 15.68 %, 18.76 % and 21.18 %, better than the benchmarked approaches. On the other hand, the proposed KMAORDM under different data packet size confirmed minimized routing overhead of 16.94 %, 19.16 % and 22.32 %, better than the baseline CORFA, GWO-MHRLM and LBAFP approaches. Moreover, the network latency incurred by the proposed KMAORDM under different data packet size is improved by 15.42 %, 18.54 % and 21.16 %, better than the benchmarked approaches.

6. Limitations

. This proposed KMAORDM scheme is determined to be more ideal for scenarios with city traffic in which adequate number of vehicles always exists in the road segment. This limitation clearly proves that the proposed KMAORDM scheme is not ideal for sparse vehicular network. It cannot achieve reliable data delivery between the source to destination under the impact of non-line of sight conditions of vehicular nodes. This approach failed to include a probabilistic data structure which may further improve the possibility of achieving improved cache strategy during the process of data delivery.

7. Future scope of enhancement

This proposed KMAORDM scheme as the part of future scope of research can be improved with the capability of handling a greater number of nodes more than 500 by adopting the impactful factors of bandwidth efficiency, packet loss and error in transmission into account. It can be improved in the near future by hybridizing the well-balanced exploration and exploitation algorithm together to achieve better solution diversity and prevent the problem of premature convergence during the research of cluster optimization in VANETs. It can be adapted into a bio-inspired clustering approach-based intrusion detection system which could handle the impact of malicious and selfish vehicular nodes in VANETs. It need to include the security dimensions associated with the routing of VANETs to handle the possible degree of threats that need to be thwarted for guaranteeing the process of reliable data transmission. Further, a hyper spectral clustering scheme which employed the matrix of modularity on par with the adjacency matrix is used for attaining effective and efficient clustering of vehicles. This proposed clustering approach can be furthermore improved with the lane index associated with the vehicles for the objective of improving the stability

of the CHs elected in the network. This proposed KMAORDM scheme can be integrated with the security mechanisms of homomorphic encryption algorithms for preventing data and vehicle authentication attack in VANETs. This proposed clustering scheme can be explored with different conditions of dynamic traffic related to sparse and dense network. This proposed KMAORDM scheme can be integrated with Internet of Things (IoT) for implementing a Internet of Vehicles environment.

CRediT authorship contribution statement

Balasubramanian C: Formal analysis, Methodology, Validation, Visualization, Writing – review & editing. **Soundarayaa R K:** Conceptualization, Data curation, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

References

[1] A. Ali, F. Aadil, M.F. Khan, M. Maqsood, S. Lim, Harris Hawks optimization-based clustering algorithm for vehicular ad-hoc networks, IEEE Trans. Intell. Transp. Syst. (2023).

[2] A. Naeem, M. Rizwan, S. Alsubai, A. Almadhor, M. Akhtaruzzaman, S. Islam, H. Rahman, Enhanced clustering based routing protocol in vehicular ad-hoc networks, IET Electr. Syst. Transp. 13 (1) (2023) e12069.

[3] J. Tiwari, A. Prakash, R. Tripathi, An adaptive and cooperative MAC protocol for safety applications in cognitive radio enabled vehicular ad-hoc networks, Ad Hoc Netw. 138 (2023) 103019.

[4] S. Ghosh, I. Saha Misra, T. Chakraborty, Improved quality of service by canine olfactory route finding algorithm for vehicular ad hoc network, Trans. Emerg. Telecommun. Technol. 34 (6) (2023) e4764.

[5] S.M. Udhaya Sankar, N. Jagadish Kumar, G. Elangovan, R. Praveen, An integrated Z-number and DEMATEL-based cooperation enforcement scheme for thwarting malicious nodes in MANETs, Wirel. Pers. Commun. 130 (4) (2023) 2531–2563.

[6] L. Rui, Z. Yan, Z. Tan, Z. Gao, Y. Yang, X. Chen, H. Liu, An intersection-based QoS routing for vehicular ad hoc networks with reinforcement learning, IEEE Trans. Intell. Transp. Syst. (2023).

[7] S. Ullah, G. Abbas, M. Waqas, Z.H. Abbas, A.U. Khan, RSU assisted reliable relay selection for emergency message routing in intermittently connected VANETs, Wirel. Netw. 29 (3) (2023) 1311–1332.

[8] B. Bhabani, J. Mahapatro, CluRMA: a cluster-based RSU-enabled message aggregation scheme for vehicular ad hoc networks, Veh. Commun. 39 (2023) 100564.

[9] S. Jagatheswari, P. Ramalingam, J. Chandra Priya, Improved grey relational analysis-based TOPSIS method for cooperation enforcing scheme to guarantee quality of service in MANETs, Int. J. Inf. Technol. 14 (2) (2022) 887–897.

[10] M. Sindhwani, S. Sachdeva, K. Arora, T. Yoon, D. Yoo, G.P. Joshi, W. Cho, Soft computing techniques aware clustering-based routing protocols in vehicular ad hoc networks: a review, Appl. Sci. 12 (15) (2022) 7922.

[11] Udhaya Sankar, R. Praveen, R. Praveen, N. Jagadish Kumar, S. Jagatheswari, Fuzzy ELECTRE multi-criteria decision-making technique for achieving reliable data dissemination in MANETs, Int. J. Inf. Technol. 15 (2023) 1985–1998.

[12] R.K. Godi, S. Janakiraman, Border Collie optimization algorithm-based node clustering technique in vehicular ad hoc networks, Trans. Emerg. Telecommun. Technol. 33 (5) (2022) e4457.

[13] T. Sudhakar, P. Ramalingam, S. Jagatheswari, An improved proxy-vehicle based authentication scheme for vehicular ad-hoc networks, Int. J. Inf. Technol. 14 (5) (2022) 2441–2449.

[14] C.J. Joshua, P. Jayachandran, A.Q. Md, A.K. Sivaraman, K.F. Tee, Clustering, routing, scheduling, and challenges in bio-inspired parameter tuning of vehicular ad hoc networks for environmental sustainability, Sustainability 15 (6) (2023) 4767.

[15] Salim, A., Khedr, A.M., Alwasel, B., Osamy, W., & Aziz, A. (2023). SOMACA: A New Swarm Optimization-Based and Mobility-Aware Clustering Approach for the Internet of Vehicles. IEEE Access.

[16] N.S. Rajput, R. Banerjee, D. Sanghi, G. Santhanam, K. Singhal, Swarm intelligence inspired meta-heuristics for solving multi-constraint QoS path problem in vehicular ad hoc networks, Ad Hoc Netw. 123 (2021) 102633.

[17] V. Ramasamy, J. Srirangan, P. Ramalingam, Fuzzy and position particle swarm optimized routing in VANET, Int. J. Electr. Comput. Eng. Syst. 12 (4) (2021) 199–206.

[18] R. Ramamoorthy, M. Thangavelu, An enhanced hybrid ant colony optimization routing protocol for vehicular ad-hoc networks, J. Ambient Intell. Humaniz. Comput. (2021) 1–32.

[19] R. Attia, A. Hassaan, R. Rizk, Advanced greedy hybrid bio-inspired routing protocol to improve IoV, IEEE Access 9 (2021) 131260–131272.

[20] Kumbhar, F.H., & Shin, S.Y. (2022). Innovating Multi-Objective Optimal Message Routing for Unified High Mobility Networks. IEEE Transactions on Vehicular Technology.

[21] R. Ramamoorthy, M. Thangavelu, An enhanced distance and residual energy-based congestion aware ant colony optimization routing for vehicular ad hoc networks, Int. J. Commun. Syst. 35 (11) (2022) e5179.

[22] R. Ramamoorthy, M. Thangavelu, An enhanced bio-inspired routing algorithm for vehicular ad hoc networks, Trends Sci. 19 (10) (2022), 4188-4188.

[23] Ishita Seth, Kalpna Guleria, Surya Narayan Panda, A lane-based advanced forwarding protocol for internet of vehicles, Int. J. Pervasive Comput. Commun. (2023).

[24] A. Nahar, D. Das, MetaLearn: optimizing routing heuristics with a hybrid meta-learning approach in vehicular ad-hoc networks, Ad Hoc Netw. 138 (2023) 102996.

[25] S. Ghosh, I. Saha Misra, T. Chakraborty, Improved quality of service by canine olfactory route-finding algorithm for vehicular ad hoc network, Trans. Emerg. Telecommun. Technol. 34 (6) (2023) e4764.

[26] K. Mershad, H. Artail, M. Gerla, ROAMER: roadside units as message routers in VANETs, Ad Hoc Netw. 10 (3) (2012) 479–496.

[27] S.S. Musa, M. Zennaro, M. Libsie, E. Pietrosemoli, Mobility-aware proactive edge caching optimization scheme in information-centric iov networks, Sensors 22 (4) (2022) 1387.

[28] H. Rana, P. Thulasiraman, R.K. Thulasiram, MAZACORNET: mobility aware zone based ant colony optimization routing for VANET. 2013 IEEE congress on evolutionary computation, IEEE,, 2013, pp. 2948–2955.

[29] H. Menouar, M. Lenardi, F. Filali, Improving proactive routing in VANETs with the MOPR movement prediction framework. 2007 7th International Conference on ITS Telecommunications, IEEE,, 2007, pp. 1–6.

[30] M. Killat, H. Hartenstein, An empirical model for probability of packet reception in vehicular ad hoc networks, EURASIP J. Wirel. Commun. Netw. 2009 (2009) 1–12.

[31] M. Ghasemi, M. Zare, A. Zahedi, M.A. Akbari, S. Mirjalili, L. Abualigah, Geyser inspired algorithm: a new geological-inspired meta-heuristic for real-parameter and constrained engineering optimization, J. Bionic Eng. (2023) 1–35.

[32] A.E. Ezugwu, J.O. Agushaka, L. Abualigah, S. Mirjalili, A.H. Gandomi, Prairie dog optimization algorithm, Neural Comput. Appl. 34 (22) (2022) 20017–20065.

[33] J.O. Agushaka, A.E. Ezugwu, L. Abualigah, Dwarf mongoose optimization algorithm, Comput. Methods Appl. Mech. Eng. 391 (2022) 114570.

[34] J.O. Agushaka, A.E. Ezugwu, L. Abualigah, Gazelle optimization algorithm: a novel nature-inspired metaheuristic optimizer, Neural Comput. Appl. 35 (5) (2023) 4099–4131.

[35] G. Hu, Y. Zheng, L. Abualigah, A.G. Hussien, DETDO: an adaptive hybrid dandelion optimizer for engineering optimization, Adv. Eng. Inform. 57 (2023) 102004.

[36] M. Zare, M. Ghasemi, A. Zahedi, K. Golalipour, S.K. Mohammadi, S. Mirjalili, L. Abualigah, A global best-guided firefly algorithm for engineering problems, J. Bionic Eng. (2023) 1–30.

[37] Abualigah, L., Ekinci, S., Izci, D., & Zitar, R.A. (2023). Modified elite opposition-based artificial hummingbird algorithm for designing FOPID controlled cruise control system. Intelligent Automation & Soft Computing.

[38] S. Khaleghian, H. Neema, M. Sartipi, T. Tran, R. Sen, A. Dubey, Calibrating real-world city traffic simulation model using vehicle speed data. 2023 IEEE International Conference on Smart Computing (SMARTCOMP), IEEE,, 2023, pp. 303–308.

[39] Rababah, M., Maydanchi, M., Pouya, S., Basiri, M., Azad, A.N., Haji, F., & Aminjarahi, M. (2022). Data Visualization of Traffic Violations in Maryland, US. arXiv preprint arXiv:2208.10543.

[40] H.Q. Yang, Z. Li, T.Q. Jie, Z.Q. Zhang, Effects of joints on the cutting behavior of disc cutter running on the jointed rock mass, Tunn. Undergr. Space Technol. 81 (2018) 112–120.

[41] H.Q. Yang, S.G. Xing, Q. Wang, Z. Li, Model test on the entrainment phenomenon and energy conversion mechanism of flow-like landslides, Eng. Geol. 239 (2018) 119–125.

[42] H.Q. Yang, Y.Y. Zeng, Y.F. Lan, X.P. Zhou, Analysis of the excavation damaged zone around a tunnel accounting for geostress and unloading, Int. J. Rock. Mech. Min. Sci. 69 (2014) 59–66.

[43] S. Madhavi, R. Praveen, N. Jagadish Kumar, S.M. Udhaya Sankar, Hybrid Grey PIPRECIA and Grey OCRA method-based dynamic multi-criteria decision-making model for mitigating non-cooperating node attacks in WSNs, Peer Peer Netw. Appl. 16 (5) (2023) 2607–2629.

[44] S. Madhavi, N.C. Santhosh, S. Rajkumar, R. Praveen, Pythagorean Fuzzy Sets-based VIKOR and TOPSIS-based multi-criteria decision-making model for mitigating resource deletion attacks in WSNs, J. Intell. Fuzzy Syst. 44 (6) (2023) 9441–9459.